New Habits, New Harms: Pediatric Dentists and Substance-Related Oral Changes in Adolescents

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Summary

In this study, we analyzed data in several categories within residents and dentists, then compared the groups' answer selections. We looked at comfort levels, clinical practices, knowledge, and quiz results between residents and dentists to attempt to discern whether there is a difference in residents and dentist responses to each group of questions. Then, we looked at the residency program for each respondent (current and past residents) to see if that informed differences in responses, and then within practicing dentists, analyzed whether tenure in their practice contributed to different responses. Finally, additional analysis questions were conducted to find out how familiarity with guidelines contributed to findings of substance use in practice, and other comparisons along those lines. All significant tests will be reported in the significant findings section, and their significance before and after adjustment will be noted.

Methods

For our methodology, we conducted Fisher exact tests, which can calculate exact probabilities to see if there's a non-random association between groups. A low p-value (after adjustment) translates to having significant evidence for there existing some association between questions.

For questions that had multiple parts, we had to split them into individual questions (e.g. 8a, 8b, 8c…) to see if there were specific responses that were influenced by the grouping variable in question, which was the role (being a resident/dentist) in the case of cross-comparison, or information like program, experience, etc. In practice, this saw us, for example, analyzing whether directors think residents should be comfortable recognizing oral soft-tissue changes related to cigarettes, ENDS/vaping, and so on, for each part. Each sub-question statistically needs to be considered independently, which is why they are separated in the analysis.

Then, to visualize the difference, we created side-by-side bar charts which plot the distribution of question responses versus the grouping variable. For resident/dentist comparisons, for example, this shows side-by-side plots of the residents' response to a question versus the dentists'. All plots, significant or not, are in the appendix.

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.1.6
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.1     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.2.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor

Attaching package: 'zoo'


The following objects are masked from 'package:base':

    as.Date, as.Date.numeric
```

Significant Findings

Below are several tables outlining each of the categories of analysis requested.

Resident/Dentist Comfort & Clinical Practice Analysis

Below is a table comparing residents and dentists comfort levels and clinical practices.

	Question1	Question2	P_value	Significant	Test	Adj_P	Sig_Adj
1	Q8a	role	0.001	TRUE	fisher	0.0053	TRUE
2	Q8h	role	0.001	TRUE	fisher	0.0053	TRUE
3	Q9	role	0.001	TRUE	fisher	0.0053	TRUE
4	Q8k	role	0.005	TRUE	fisher	0.0200	TRUE
5	Q8g	role	0.025	TRUE	fisher	0.0667	FALSE
6	Q11a	role	0.0225	TRUE	fisher	0.0667	FALSE
7	Q8e	role	0.03	TRUE	fisher	0.0686	FALSE
8	Q8c	role	0.042	TRUE	fisher	0.0791	FALSE

9	Q8d	role	0.0445	TRUE	fisher	0.0791	FALSE
10	Q8j	role	0.066	FALSE	fisher	0.1056	FALSE
11	Q8b	role	0.1044	FALSE	fisher	0.1519	FALSE
12	Q10a	role	0.1309	FALSE	fisher	0.1745	FALSE
13	Q11b	role	0.1469	FALSE	fisher	0.1808	FALSE
14	Q10b	role	0.1644	FALSE	fisher	0.1879	FALSE
15	Q8f	role	0.1919	FALSE	fisher	0.2047	FALSE
16	Q8i	role	0.4038	FALSE	fisher	0.4038	FALSE

Significant findings include, with context:

Q8h, Q9, Q8a, and Q8k, all compared to the role (resident/dentist): This means that for each of the following questions, dentists and residents' responses differed significantly in regards to comfort level/clinical practices

Q8h: Determining when an oral soft-tissue finding warrants referral or biopsy

Q8a: Recognizing oral soft-tissue changes associated with cigarette smoking in adolescents

Q8k: Knowledge and understanding of current AAPD guidelines related to substance-use, anticipatory guidance, and management

Q9: How often do you screen patients for tobacco, vaping/ENDS, or other substance use?

Resident/Dentist Knowledge Analysis

Below is a table comparing residents and dentist questions on knowledge.

	Question1	Question2	P_value	Significant	Test	Adj_P	Sig_Adj
1	Q14	role	0.0515	FALSE	fisher	0.1545	FALSE
2	Q13	role	0.1934	FALSE	fisher	0.2901	FALSE
3	Q12	role	0.3488	FALSE	fisher	0.3488	FALSE

Out of the three questions analyzed, none of them found significance. We did not find significant evidence to suggest that residents/dentists had different responses to these questions based on knowledge.

Residents/Dentists Quiz Analysis

Below is a table comparing residents and dentist quiz results.

	Question1	Question2	P_value	Significant	Test	Adj_P	Sig_Adj
1	Q15	role	0.8836	FALSE	fisher	0.8836	FALSE
2	Q16	role	0.8151	FALSE	fisher	0.8836	FALSE
3	Q17	role	0.8101	FALSE	fisher	0.8836	FALSE
4	Q18	role	0.5337	FALSE	fisher	0.8836	FALSE
5	Q19	role	0.6377	FALSE	fisher	0.8836	FALSE
6	Q20	role	0.2199	FALSE	fisher	0.8836	FALSE
7	score	role	0.4698	FALSE	fisher	0.8836	FALSE

Similarly, out of the 7 questions analyzed, including the overall quiz score, none of them found significance. We did not find significant evidence to suggest that residents/dentists had different responses to each quiz question nor in aggregate.

Environment Analysis

Below is a table comparing environmental questions.

	Question1	Question2	P_value	Significant	Test	Adj_P	Sig_Adj
1	Q8a	Q5	0.033	TRUE	fisher	0.1034	FALSE
2	Q8c	Q5	0.0365	TRUE	fisher	0.1034	FALSE
3	Q8g	Q5	0.0105	TRUE	fisher	0.1034	FALSE
4	Q8h	Q5	0.02	TRUE	fisher	0.1034	FALSE
5	Q8k	Q5	0.0335	TRUE	fisher	0.1034	FALSE
6	Q13	Q5	0.024	TRUE	fisher	0.1034	FALSE
7	Q9	Q5	0.1794	FALSE	fisher	0.4357	FALSE
8	Q8b	Q5	0.2384	FALSE	fisher	0.4871	FALSE
9	Q8f	Q5	0.2579	FALSE	fisher	0.4871	FALSE
10	Q10a	Q5	0.2914	FALSE	fisher	0.4954	FALSE
11	Q8d	Q5	0.5542	FALSE	fisher	0.7851	FALSE
12	Q8e	Q5	0.5437	FALSE	fisher	0.7851	FALSE
13	Q8i	Q5	0.7366	FALSE	fisher	0.8348	FALSE
14	Q8j	Q5	0.7016	FALSE	fisher	0.8348	FALSE
15	Q10b	Q5	0.6537	FALSE	fisher	0.8348	FALSE
16	Q12	Q5	0.9835	FALSE	fisher	0.9915	FALSE
17	score	Q5	0.9915	FALSE	fisher	0.9915	FALSE

Out of the 17 comparisons, 6 were found significant, but none after adjusting for the amount of tests conducted. Still, the original significant results are recorded below:

Q8a, c, g, h, k, Q13 vs Q5:

a: Recognizing oral soft-tissue changes associated with cigarette smoking in adolescents

c: Recognizing oral soft-tissue changes associated with marijuana use (smoked or vaped) in adolescents

g: Identifying potentially malignant oral lesions in adolescents

h: Determining when an oral soft-tissue finding warrants referral or biopsy

k: Knowledge and understanding of current AAPD guidelines related to substance-use, anticipatory guidance, and management

13: According to AAPD guidelines, at what age should anticipatory guidance and counseling regarding tobacco vaping/ ENDS, and substance misuse begin?

All questions were analyzed compared to the residency program, showing that there is significant evidence for each of these tests, before adjusting for multiple tests, that there is a difference in responses to the above questions based on program.

Dentist/Practice Analysis

Below is a table comparing dentists and their practices.

	Question1	Question2	P_value	Significant	Test	Adj_P	Sig_Adj
1	Q9	Q3	0.0035	TRUE	fisher	0.0630	FALSE
2	score	Q3	0.0075	TRUE	fisher	0.0675	FALSE
3	Q8d	Q3	0.0235	TRUE	fisher	0.1057	FALSE
4	Q10b	Q3	0.02	TRUE	fisher	0.1057	FALSE
5	Q8a	Q3	0.036	TRUE	fisher	0.1296	FALSE
6	Q8b	Q3	0.053	FALSE	fisher	0.1590	FALSE
7	Q8f	Q3	0.074	FALSE	fisher	0.1903	FALSE
8	Q8i	Q3	0.1029	FALSE	fisher	0.2058	FALSE
9	Q12	Q3	0.0945	FALSE	fisher	0.2058	FALSE
10	Q10a	Q3	0.2614	FALSE	fisher	0.4705	FALSE
11	Q8j	Q3	0.2934	FALSE	fisher	0.4801	FALSE
12	Q8c	Q3	0.4578	FALSE	fisher	0.5494	FALSE
13	Q8e	Q3	0.4173	FALSE	fisher	0.5494	FALSE
14	Q8h	Q3	0.4508	FALSE	fisher	0.5494	FALSE
15	Q11a	Q3	0.4423	FALSE	fisher	0.5494	FALSE
16	Q8g	Q3	0.5847	FALSE	fisher	0.6191	FALSE
17	Q11b	Q3	0.5702	FALSE	fisher	0.6191	FALSE

```
18       Q8k        Q3  0.8771       FALSE fisher 0.8771   FALSE
```

Out of 18 comparisons, 6 were found to be significant, but only 1 after adjustment. All contexts are reported below:

All questions were compared to Q3, which is the amount of years a dentist has spent in their practice.

Q9 vs Q3 (only significant question after adjustment): Years in practice vs frequency of screening patients for substance abuse

Q8a, b, d: Comfort level of:

a: Recognizing oral soft-tissue changes associated with cigarette smoking in adolescents

b: Recognizing oral soft-tissue changes associated with electronic nicotine delivery systems (ENDS/vaping) in adolescents

d: Recognizing oral soft-tissue changes associated with alcohol use in adolescents

Q10b: What age do you begin:

b: Providing anticipatory guidance or counseling related to tobacco, vaping/ENDS, or substance misuse

Score: Overall quiz score

Additional Analysis

Below is a table comparing additional analysis questions.

```
   Question1 Question2 P_value Significant   Test  Adj_P Sig_Adj
1         Q8a       Q12   5e-04        TRUE fisher 0.0006    TRUE
2         Q8b       Q12   5e-04        TRUE fisher 0.0006    TRUE
3         Q8d       Q12   5e-04        TRUE fisher 0.0006    TRUE
4         Q8e       Q12   5e-04        TRUE fisher 0.0006    TRUE
5         Q8f       Q12   5e-04        TRUE fisher 0.0006    TRUE
6         Q8h       Q12   5e-04        TRUE fisher 0.0006    TRUE
7         Q8i       Q12   5e-04        TRUE fisher 0.0006    TRUE
8         Q8j       Q12   5e-04        TRUE fisher 0.0006    TRUE
9         Q8k       Q12   5e-04        TRUE fisher 0.0006    TRUE
10         Q9       Q12   5e-04        TRUE fisher 0.0006    TRUE
11       Q10a       Q12   5e-04        TRUE fisher 0.0006    TRUE
12       Q10b       Q12   5e-04        TRUE fisher 0.0006    TRUE
13       Q11a        Q9   5e-04        TRUE fisher 0.0006    TRUE
14       Q11b        Q9   5e-04        TRUE fisher 0.0006    TRUE
```

```
15      Q10a       Q13   5e-04        TRUE fisher 0.0006    TRUE
16      Q10b       Q13   5e-04        TRUE fisher 0.0006    TRUE
17       Q8g       Q12   0.001        TRUE fisher 0.0011    TRUE
18       Q8c       Q12   0.002        TRUE fisher 0.0020    TRUE
```

For additional analysis questions, surprisingly, all comparisons (18) evaluated to have significance before and after adjustment. These include:

Q8: All topics of comfort level vs Q12: Familiarity with AAPD policies

Q9: Screening frequency for substances vs Q12: Familiarity with AAPD policies

Q10a, Q10b: What age to begin screening + providing guidance for substance use vs Q12: Familiarity with AAPD policies

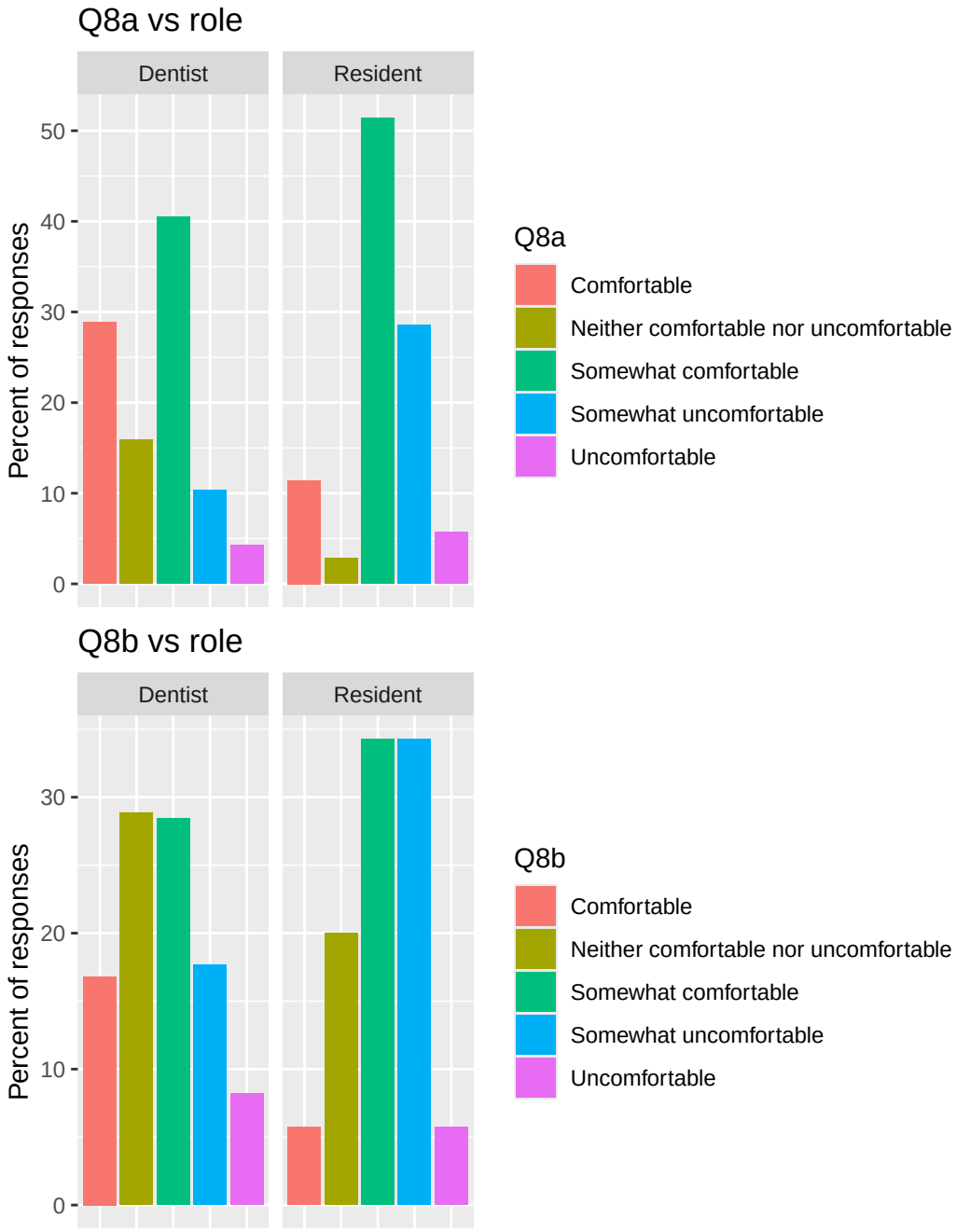
Q10a, Q10b: What age to begin screening + providing guidance for substance use vs Q13: What age should anticipatory guidance and counseling regarding tobacco vaping/ ENDS, and substance misuse begin?

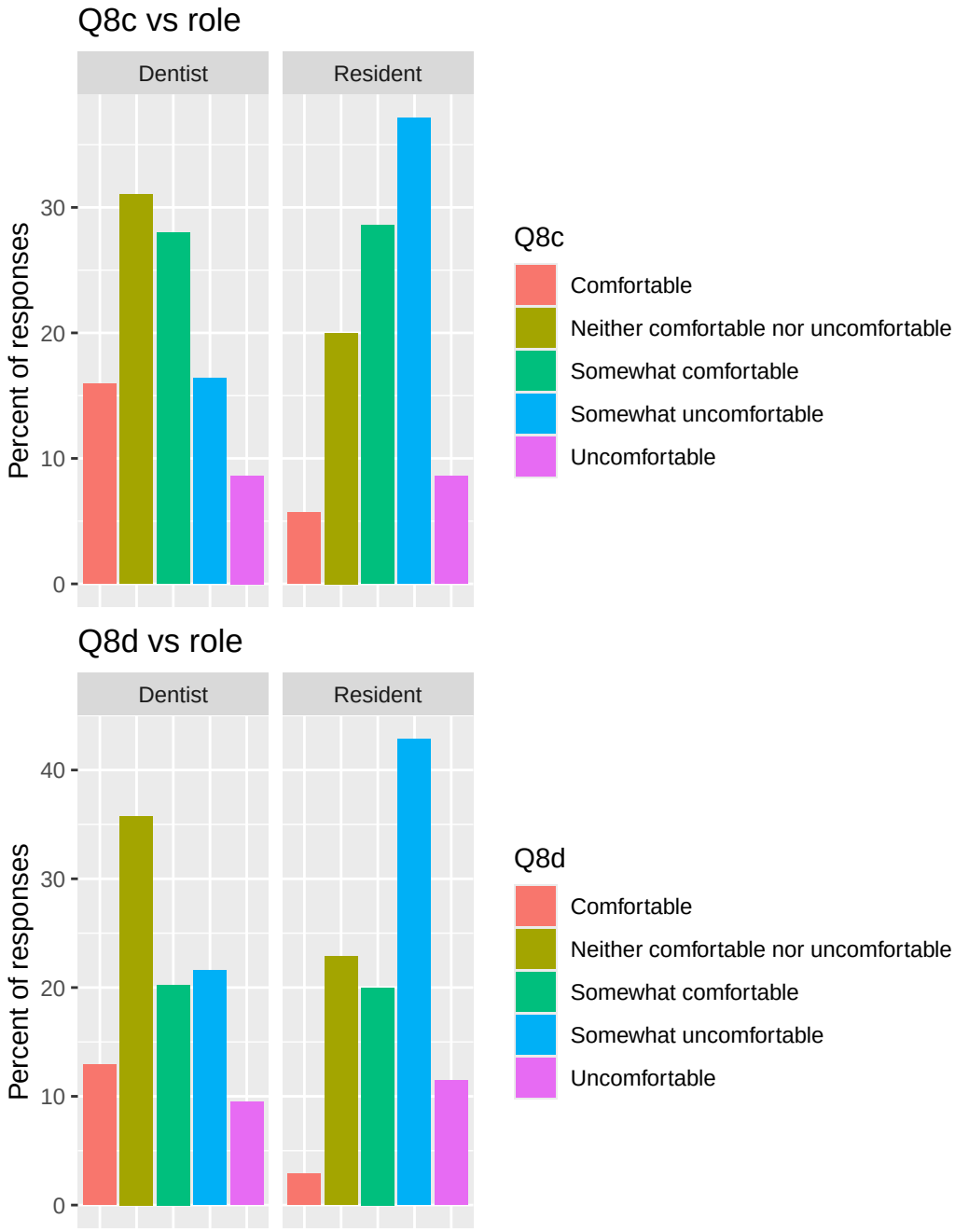
Q11a, Q11b: Substances identified through history + oral findings vs Q9: Frequency of screening patients for substance use

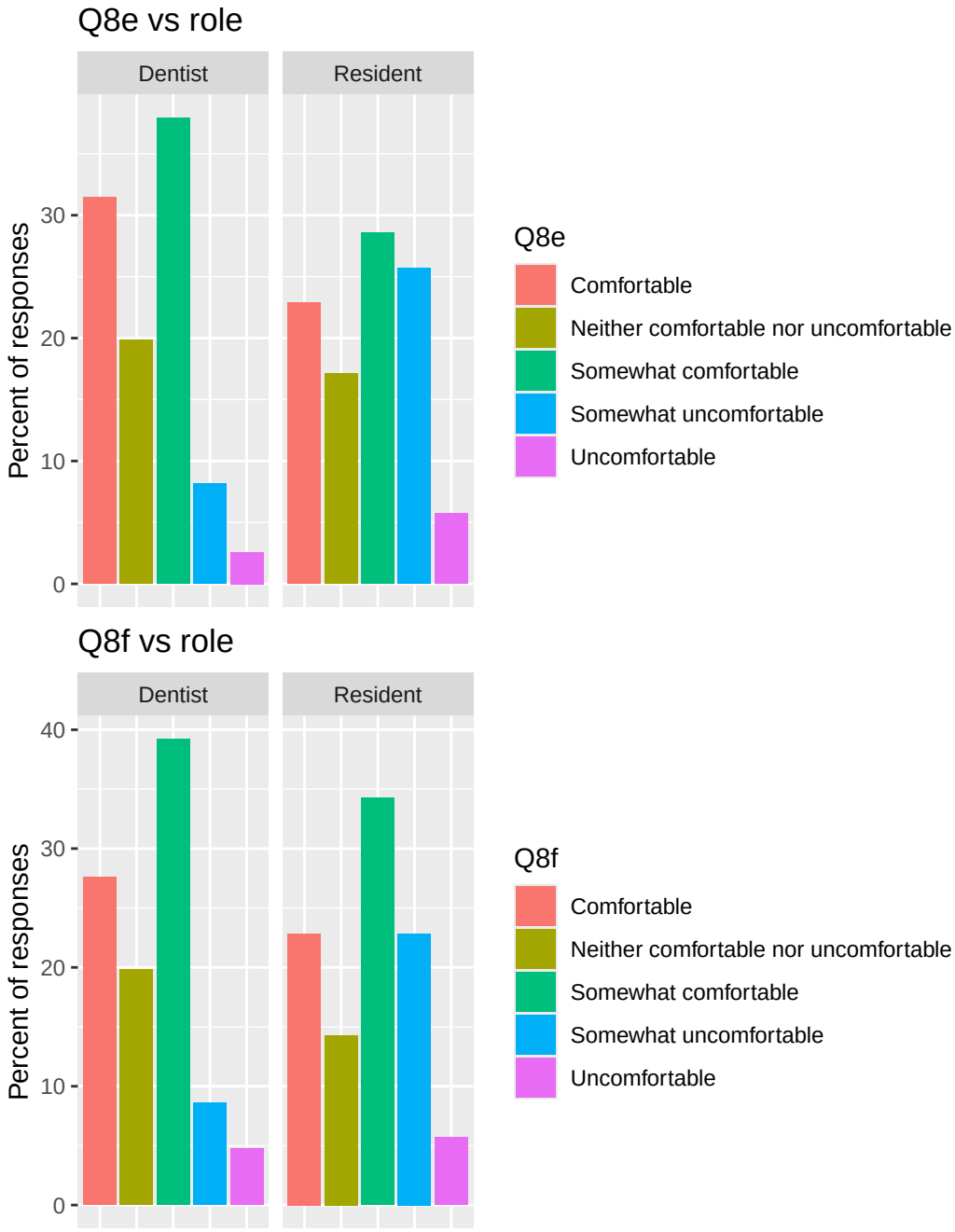
Appendix A: Plots

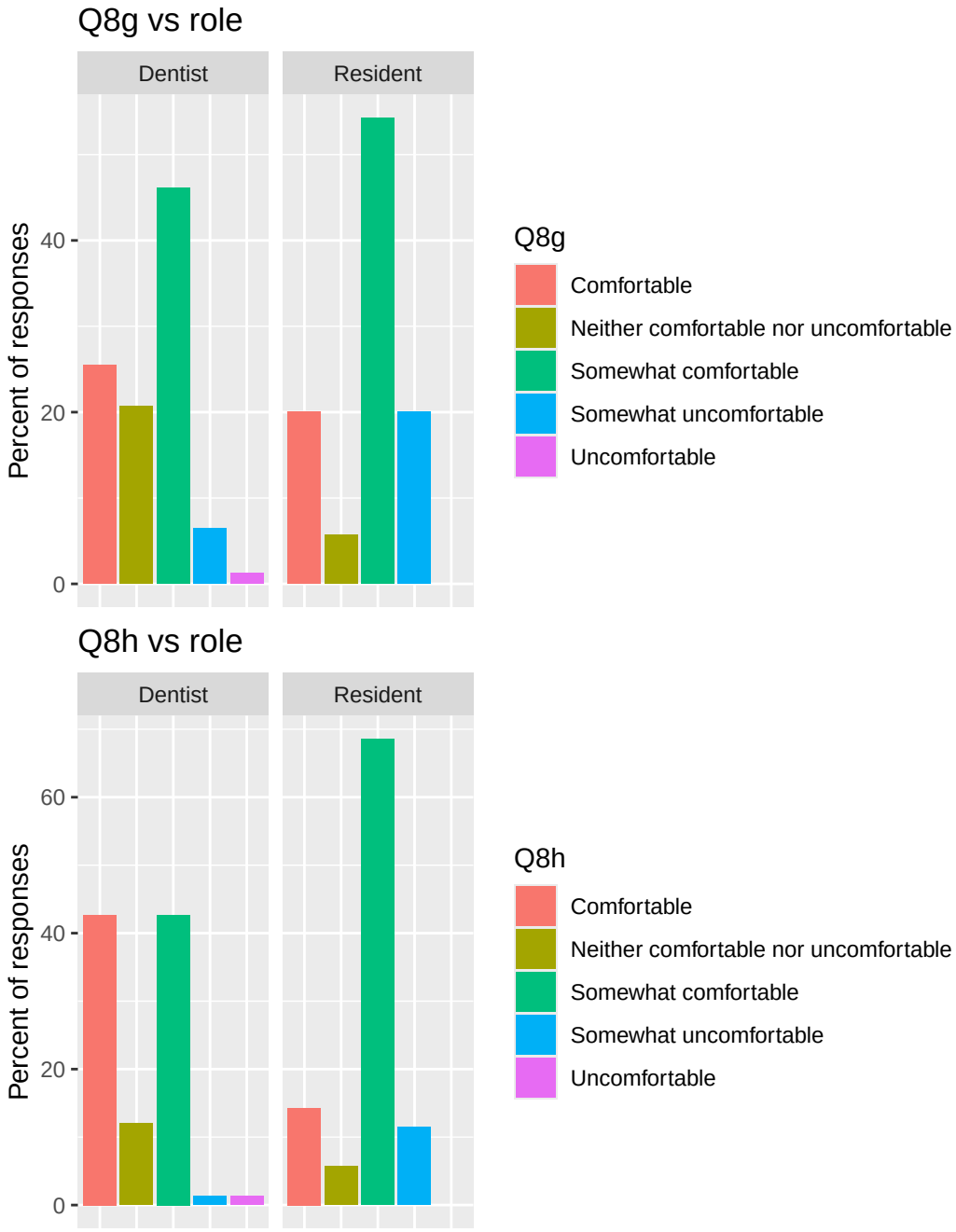
Comfort & Clinical Practice Plots

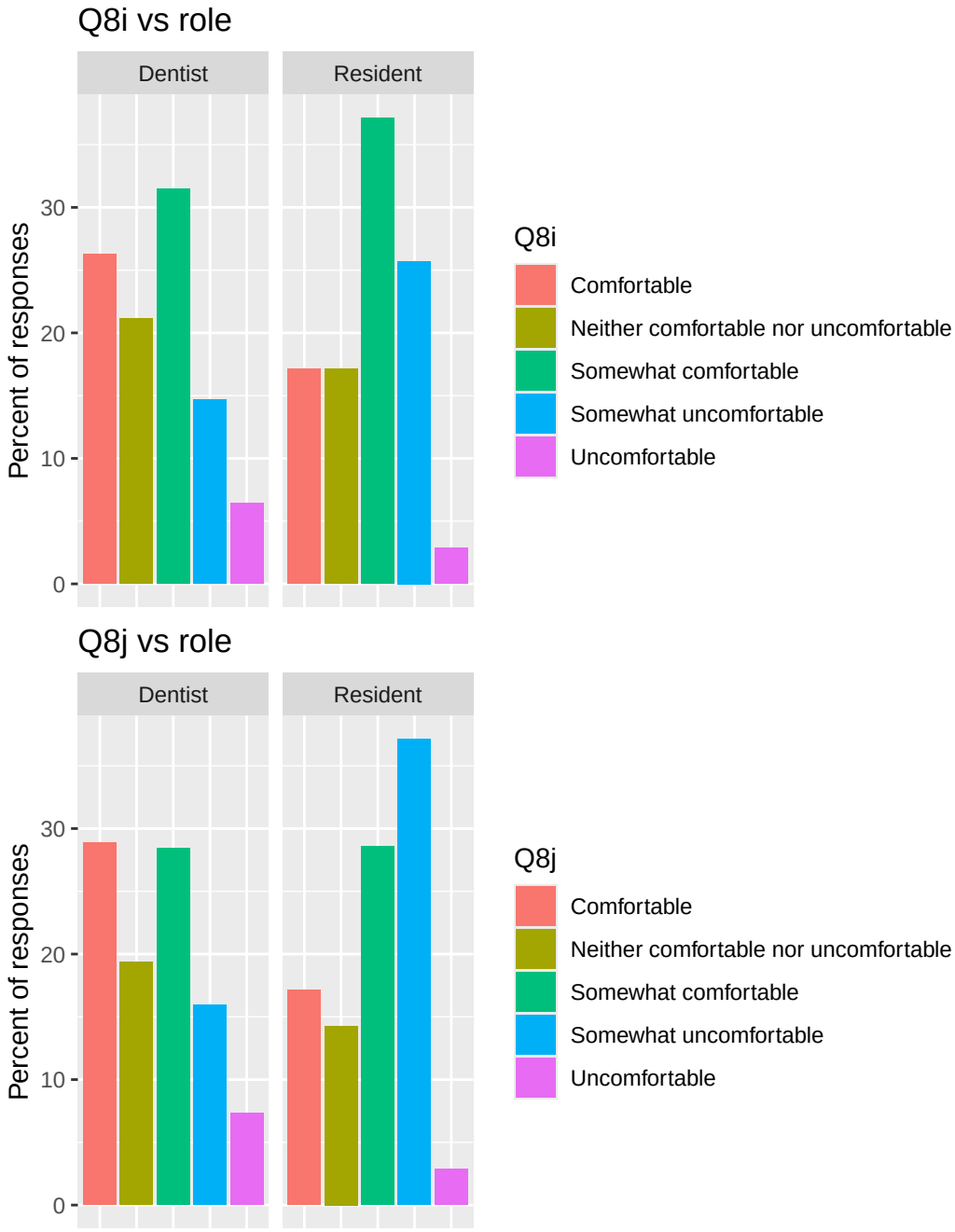
Below are all plots comparing the comfort and clinical practices between residents and dentists.

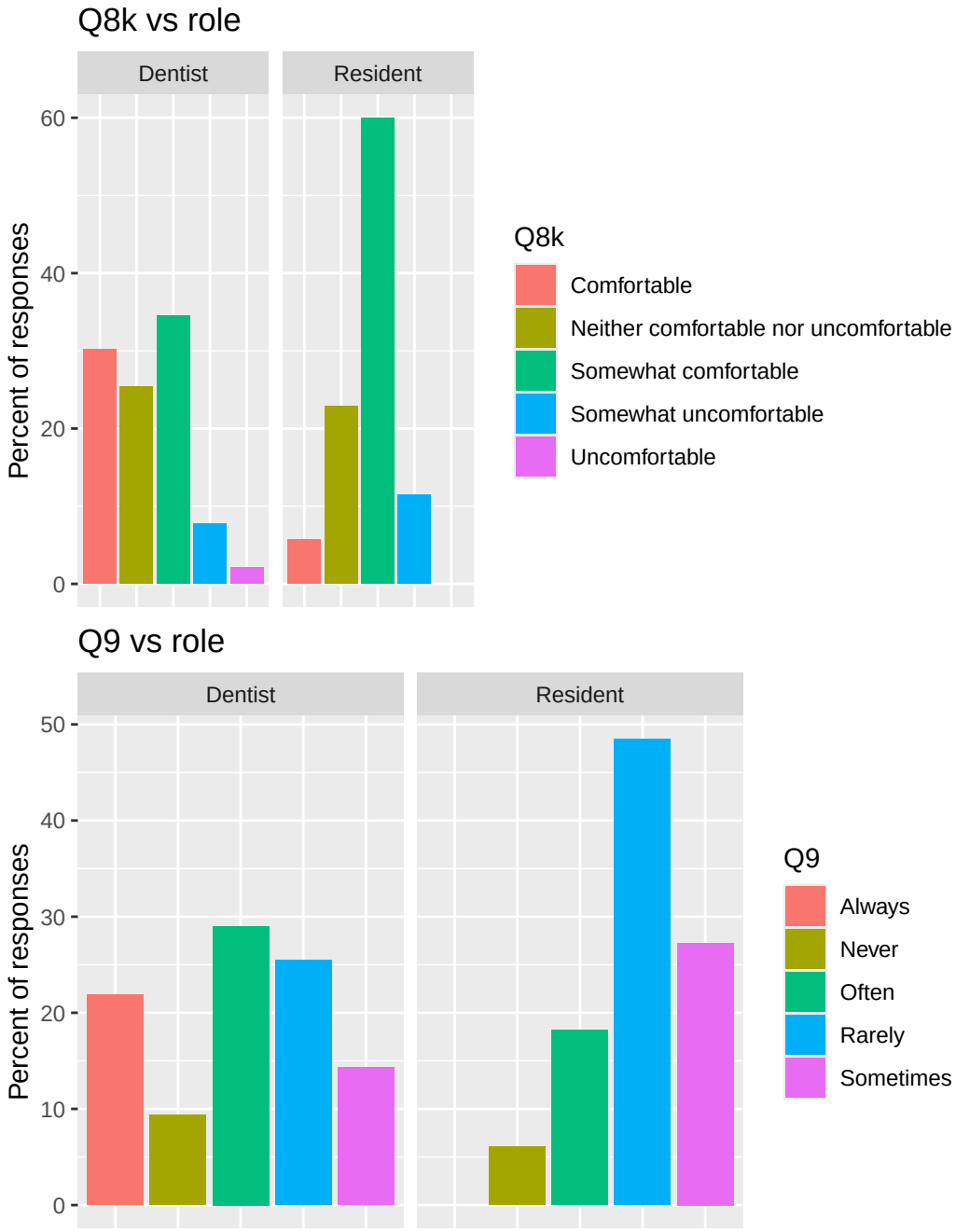

Q8a vs role
Dentist
Resident
Percent of responses
0
10
20
30
40
50
Q8a
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8b vs role
Dentist
Resident
Percent of responses
0
10
20
30
Q8b
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

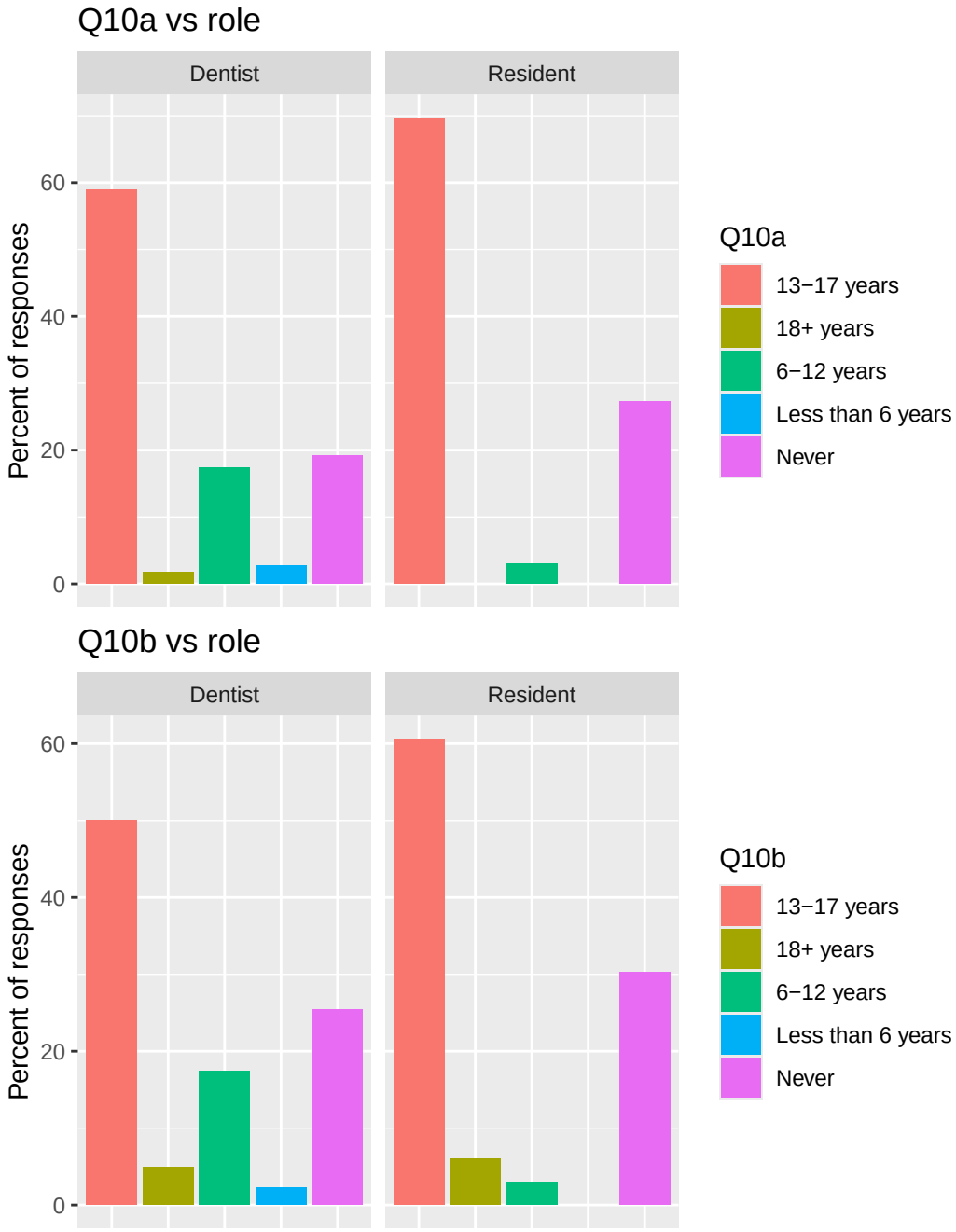

Q8c vs role
Dentist
Resident
Percent of responses
0
10
20
30
Q8c
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8d vs role
Dentist
Resident
Percent of responses
0
10
20
30
40
Q8d
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

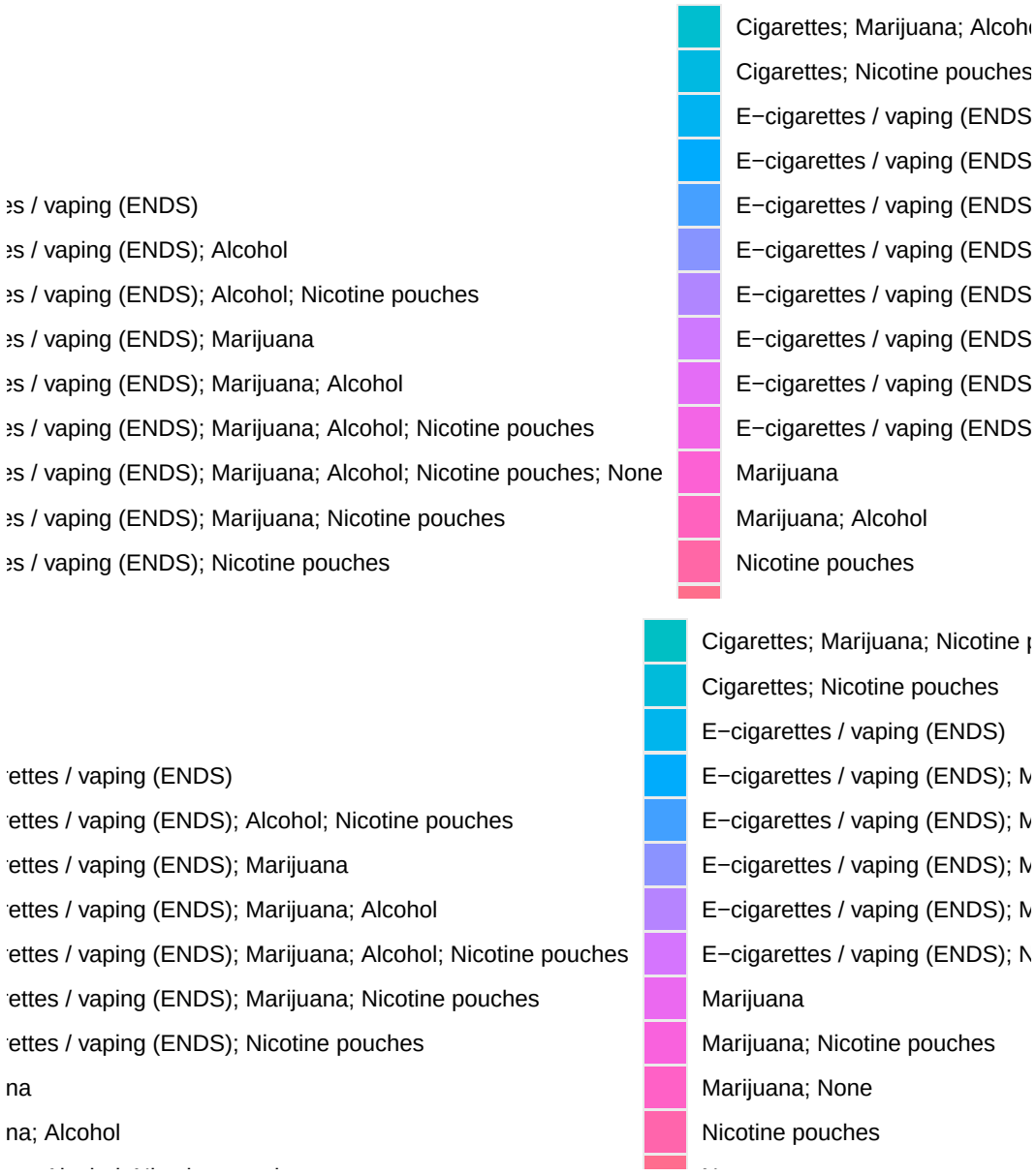

Q8e vs role
Dentist
Resident
Percent of responses
0
10
20
30
Q8e
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8f vs role
Dentist
Resident
Percent of responses
0
10
20
30
40
Q8f
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable


Q8g vs role
Dentist
Resident
Percent of responses
40
20
0
Q8g
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8h vs role
Dentist
Resident
Percent of responses
60
40
20
0
Q8h
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable


Q8i vs role
Dentist
Resident
Percent of responses
30
20
10
0
Q8i
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8j vs role
Dentist
Resident
Percent of responses
30
20
10
0
Q8j
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

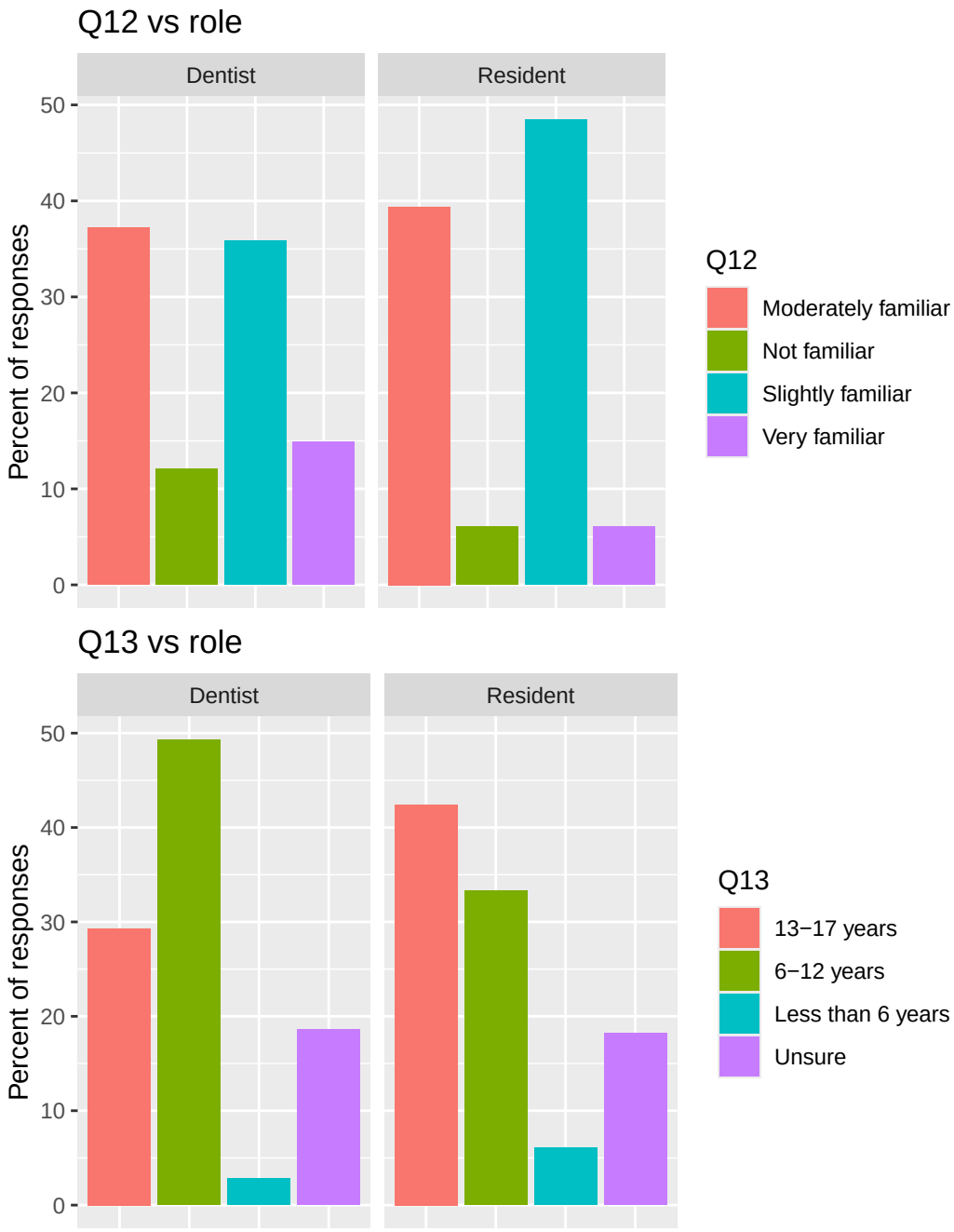

Q8k vs role
Dentist
Resident
Percent of responses
60
40
20
0
Q8k
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q9 vs role
Dentist
Resident
Percent of responses
50
40
30
20
10
0
Q9
Always
Never
Often
Rarely
Sometimes

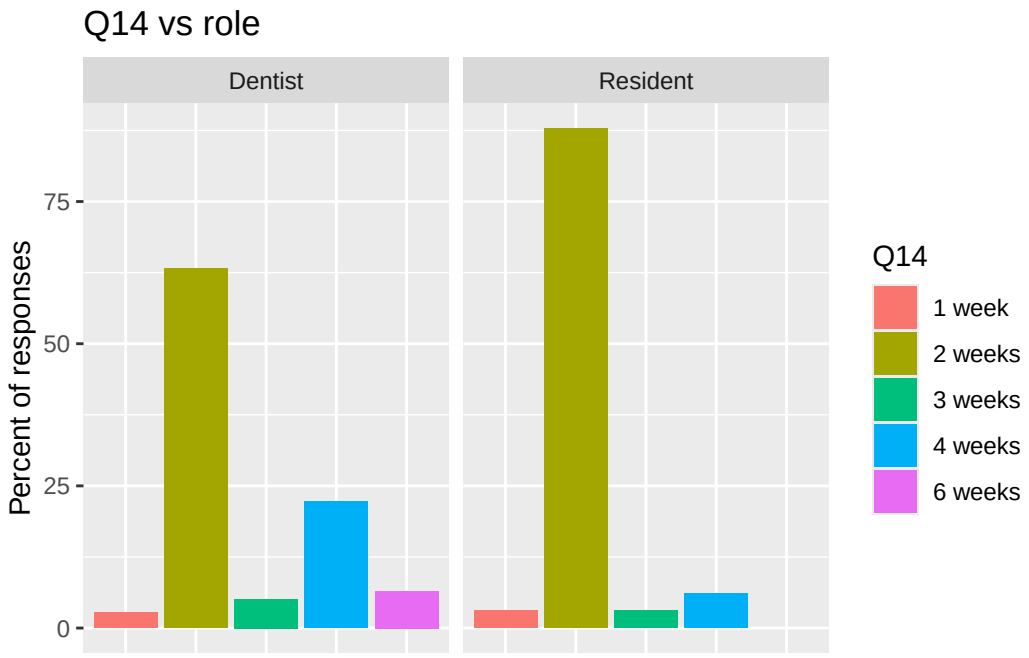

Q10a vs role
Dentist
Resident
Percent of responses
60
40
20
0
Q10a
13−17 years
18+ years
6−12 years
Less than 6 years
Never
Q10b vs role
Dentist
Resident
Percent of responses
60
40
20
0
Q10b
13−17 years
18+ years
6−12 years
Less than 6 years
Never



Knowledge Plots

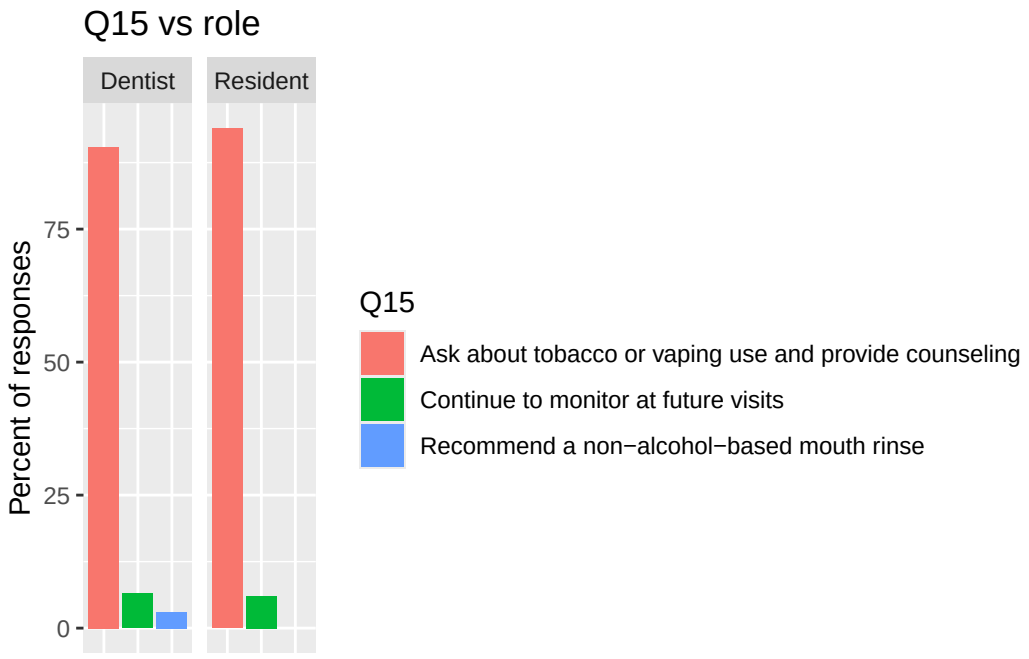
Below are all plots comparing the knowledge of residents versus dentists.

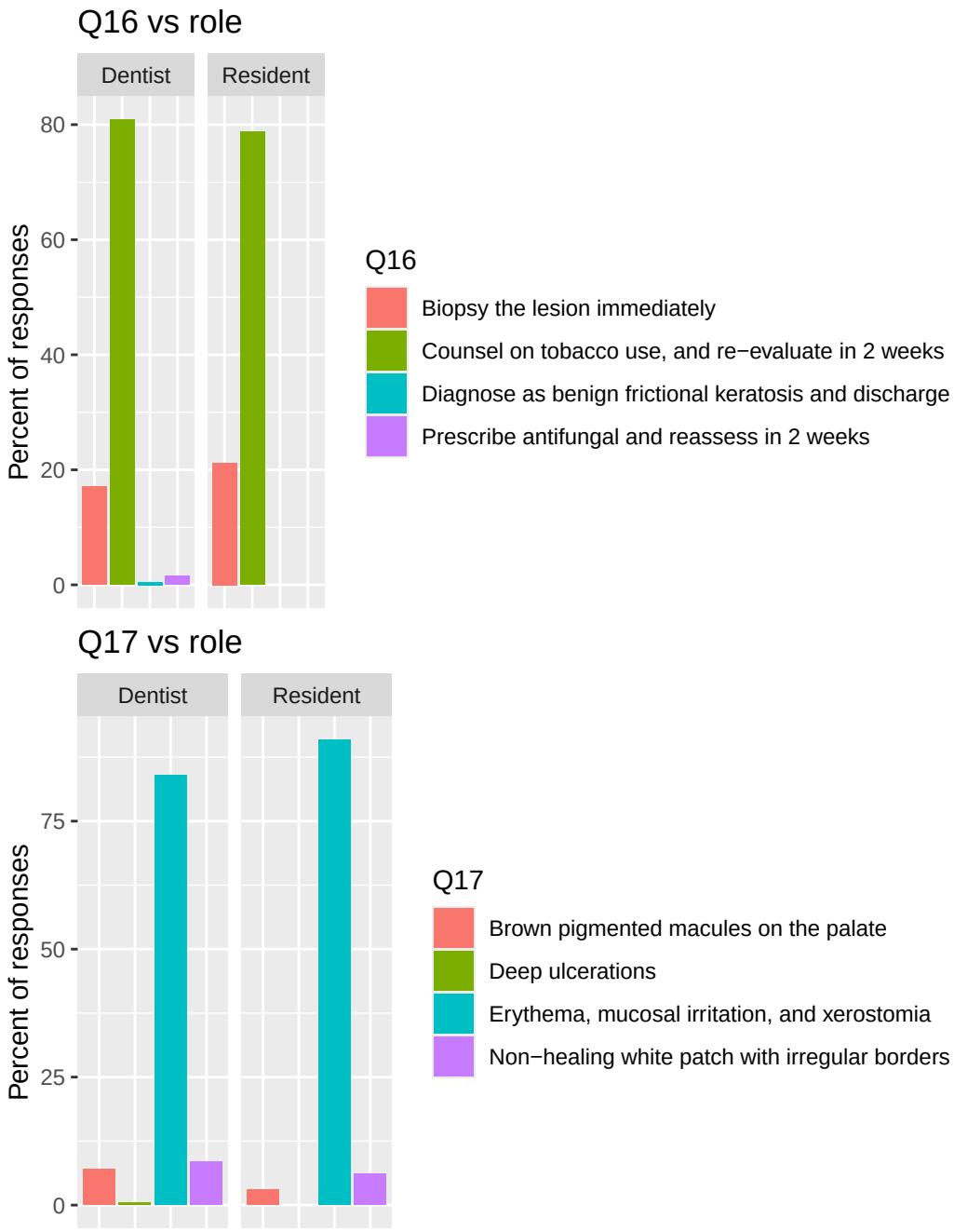

Q12 vs role
Dentist
Resident
Percent of responses
50
40
30
20
10
0
Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Q13 vs role
Dentist
Resident
Percent of responses
50
40
30
20
10
0
Q13
13−17 years
6−12 years
Less than 6 years
Unsure

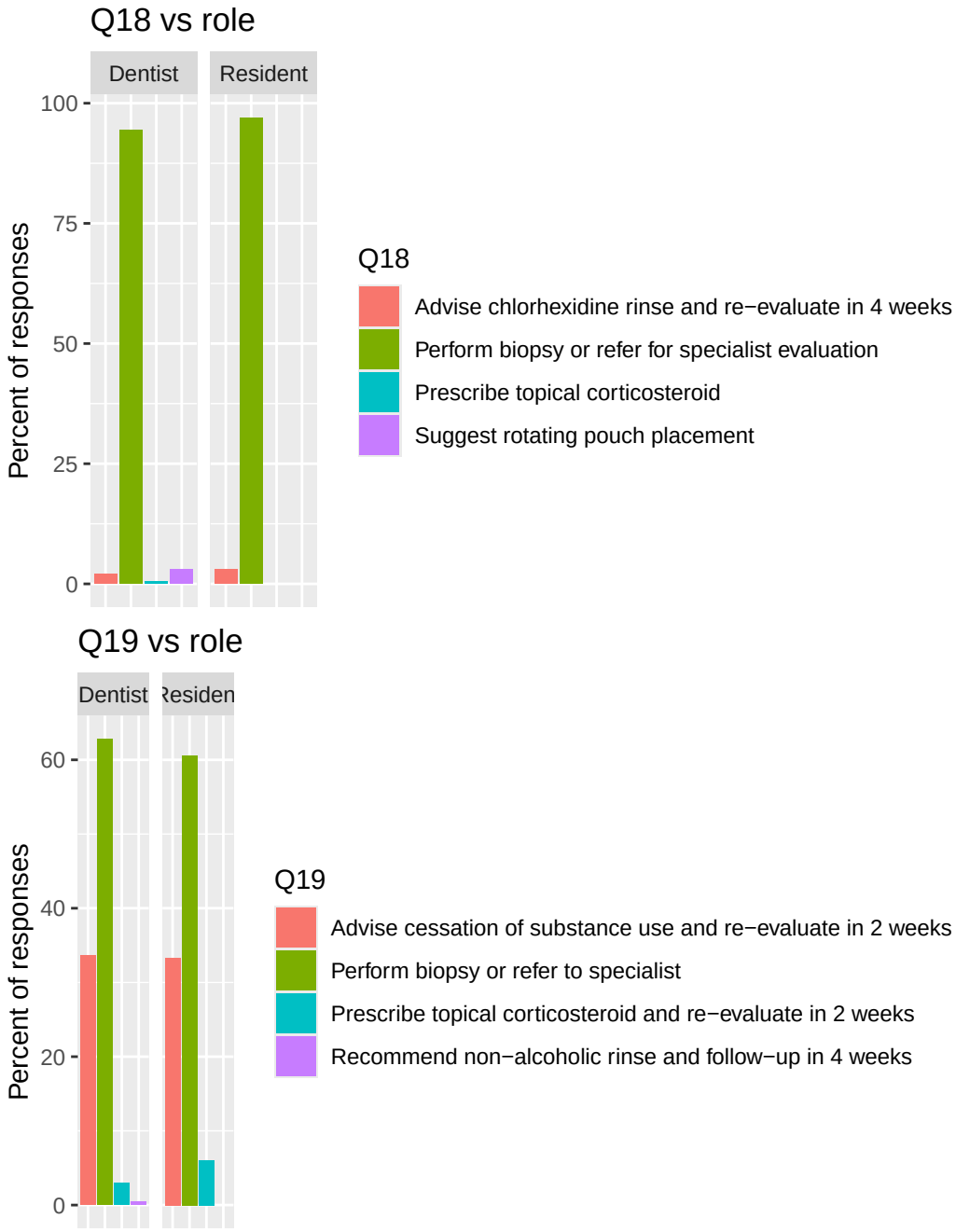


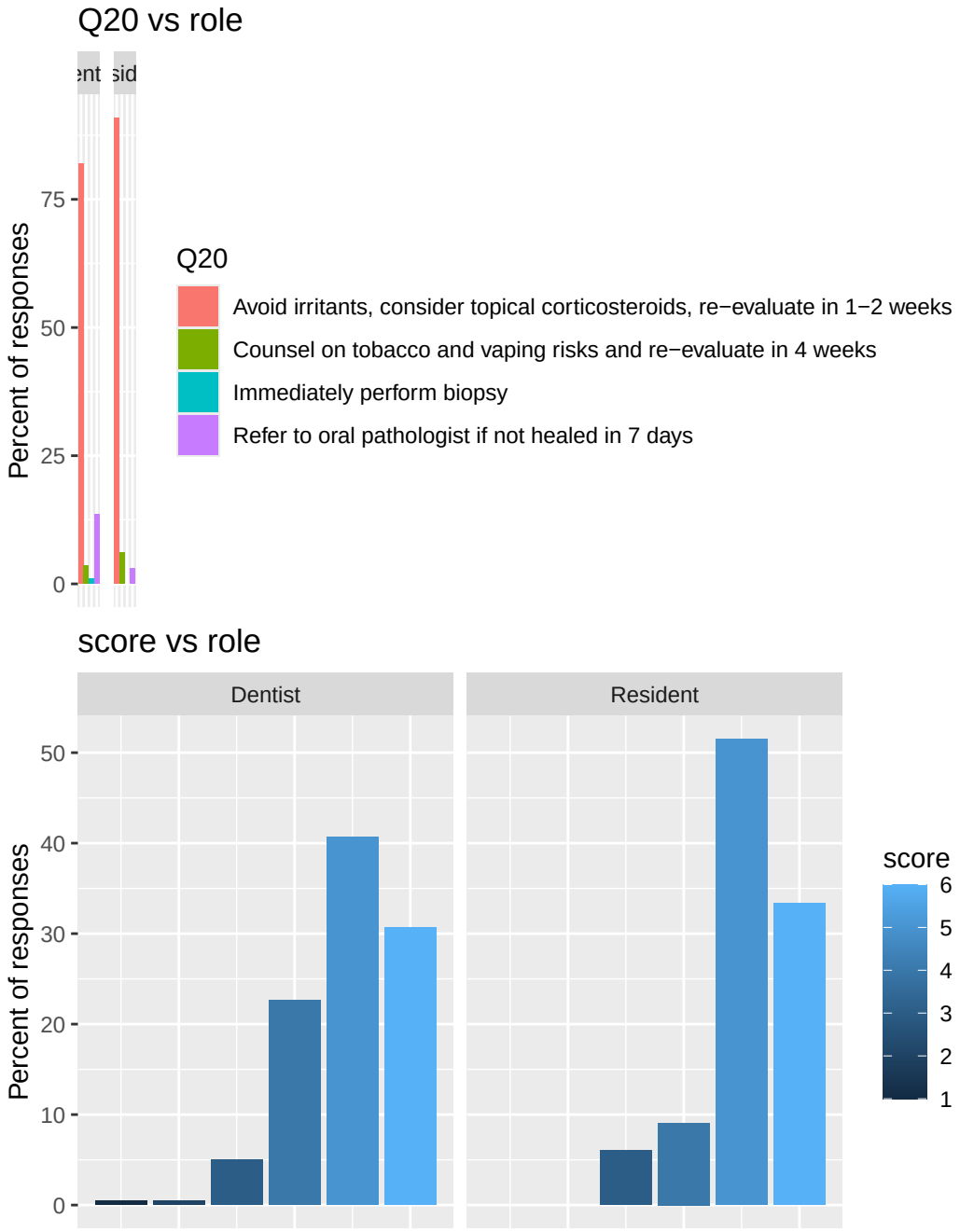
Quiz Plots

Below are all plots comparing the quiz results between residents and dentists.



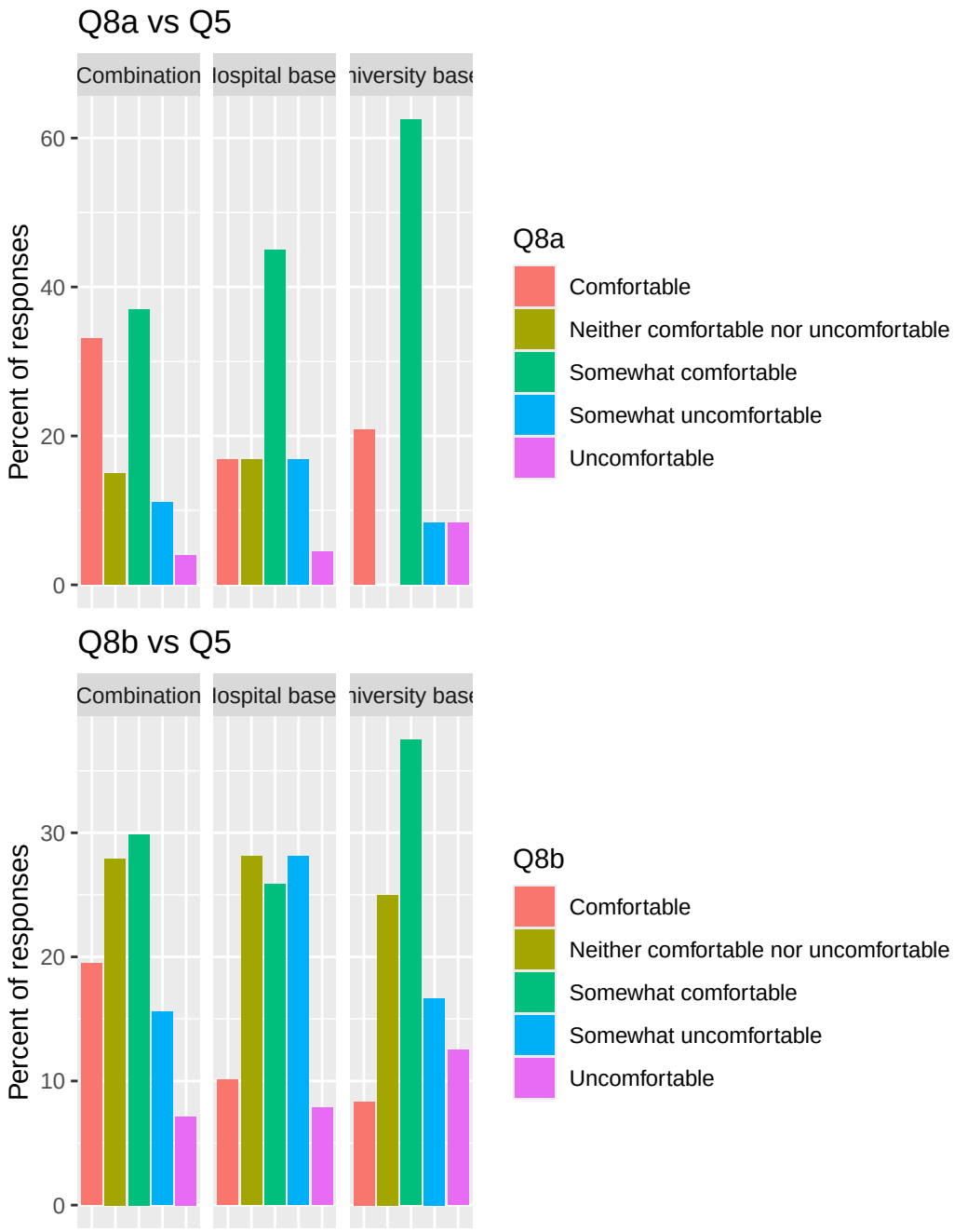

Q16 vs role
Dentist
Resident
Percent of responses
80
60
40
20
0
Q16
Biopsy the lesion immediately
Counsel on tobacco use, and re−evaluate in 2 weeks
Diagnose as benign frictional keratosis and discharge
Prescribe antifungal and reassess in 2 weeks
Q17 vs role
Dentist
Resident
Percent of responses
75
50
25
0
Q17
Brown pigmented macules on the palate
Deep ulcerations
Erythema, mucosal irritation, and xerostomia
Non−healing white patch with irregular borders

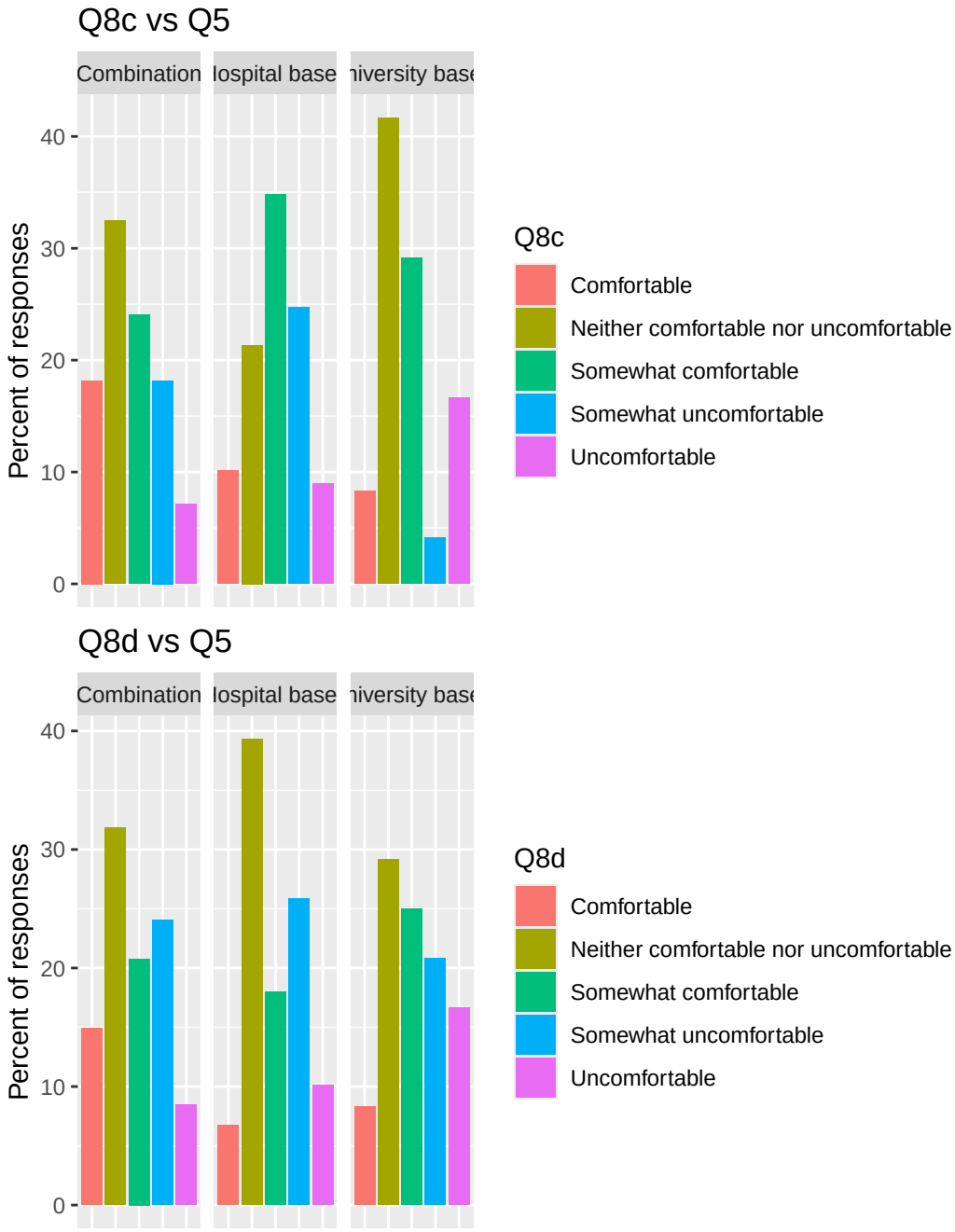

Q18 vs role
Dentist
Resident
Percent of responses
100
75
50
25
0
Q18
Advise chlorhexidine rinse and re−evaluate in 4 weeks
Perform biopsy or refer for specialist evaluation
Prescribe topical corticosteroid
Suggest rotating pouch placement
Q19 vs role
Dentist
Residen
Percent of responses
60
40
20
0
Q19
Advise cessation of substance use and re−evaluate in 2 weeks
Perform biopsy or refer to specialist
Prescribe topical corticosteroid and re−evaluate in 2 weeks
Recommend non−alcoholic rinse and follow−up in 4 weeks

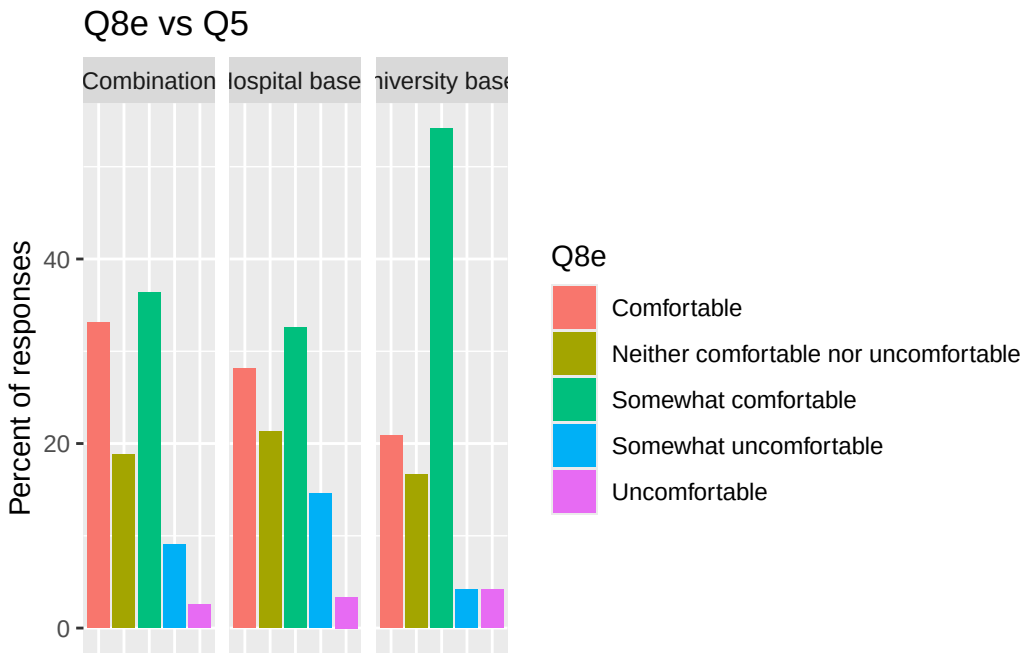


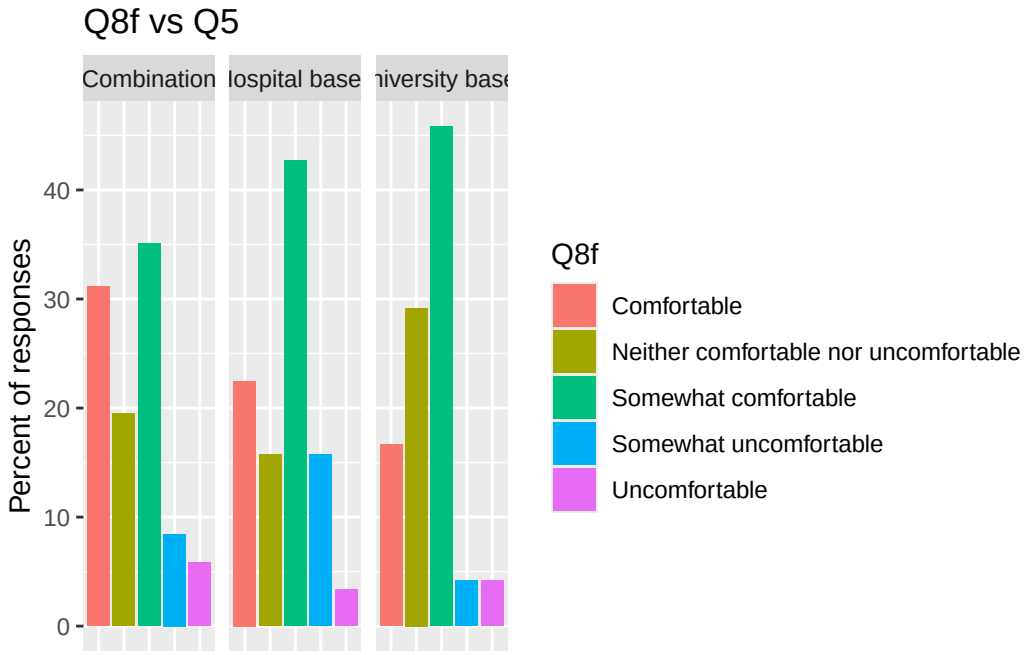
Environmental Analysis Plots

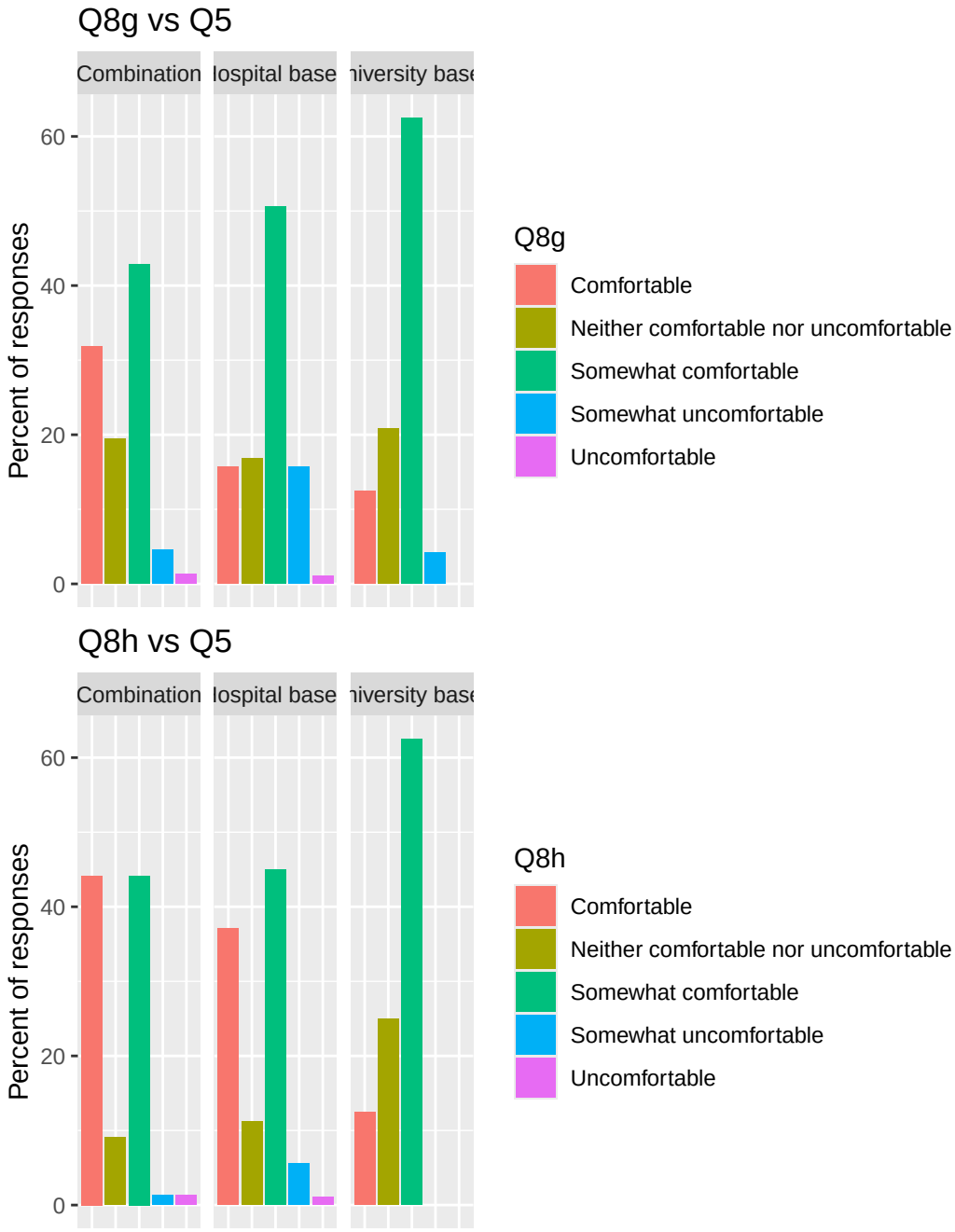
Below are plots for the environmental analysis.

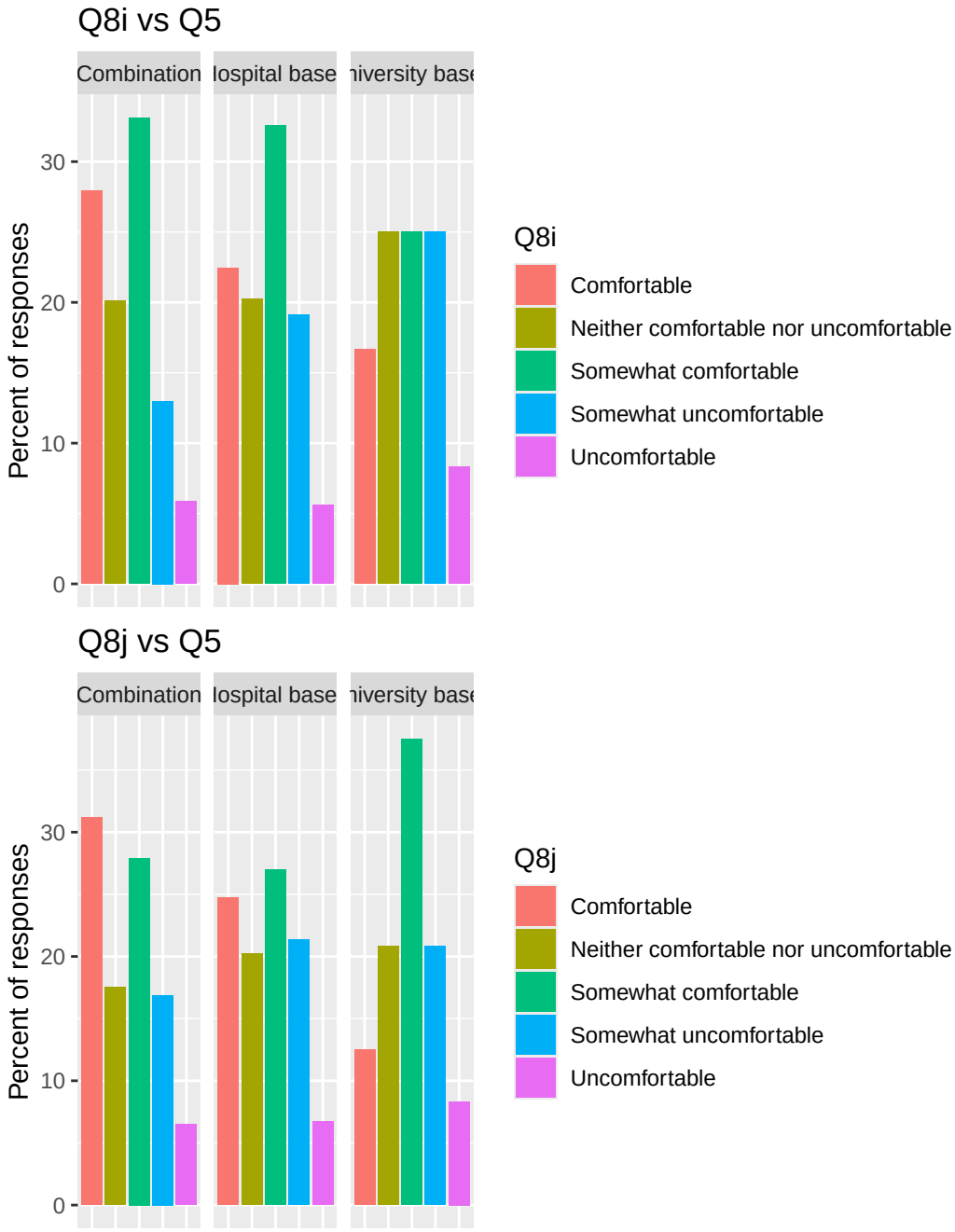

Q8a vs Q5
Combination
lospital base
niversity base
Percent of responses
0
20
40
60
Q8a
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8b vs Q5
Combination
lospital base
niversity base
Percent of responses
0
10
20
30
Q8b
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

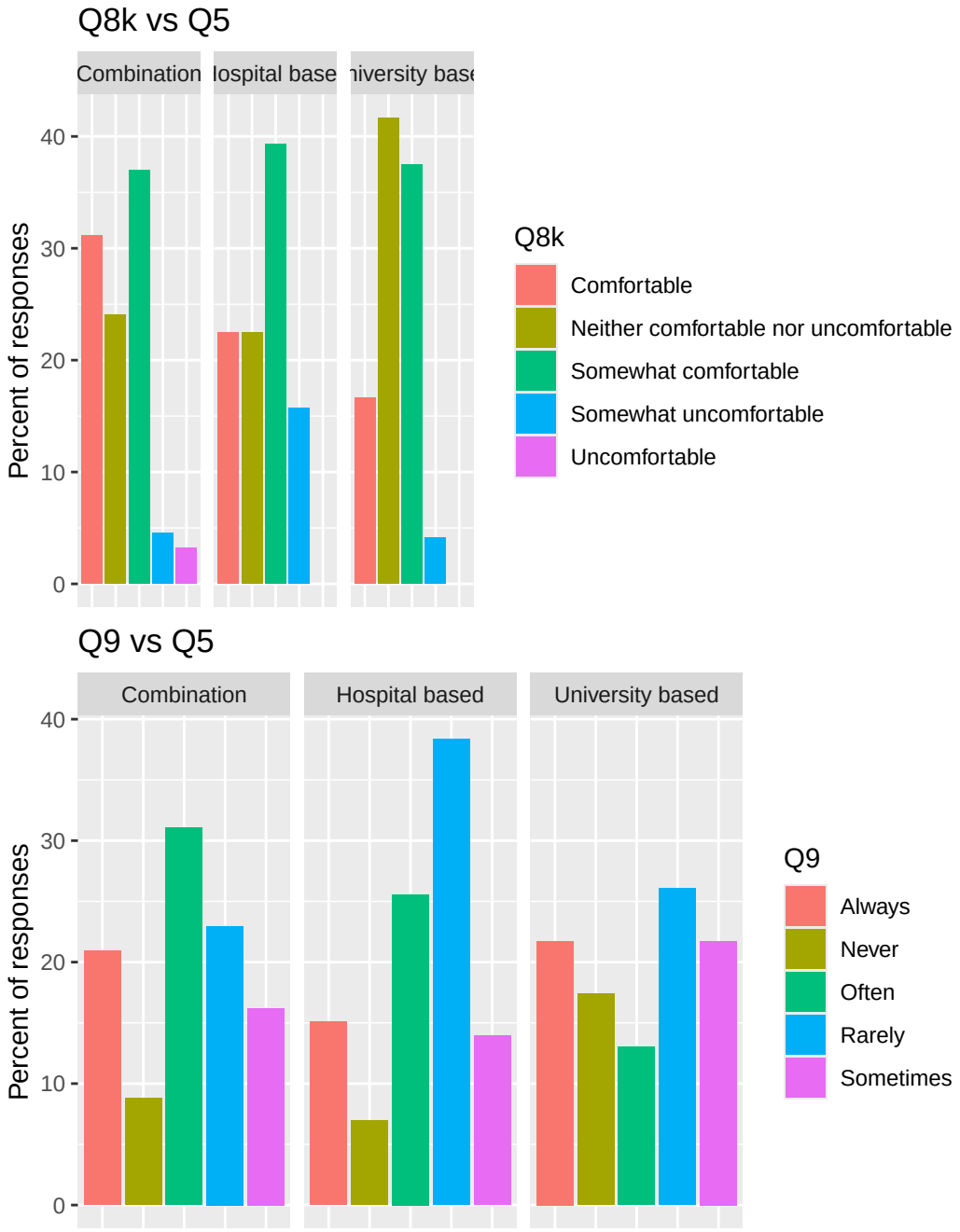

Q8c vs Q5
Combination
lospital base
niversity base
Percent of responses
0
10
20
30
40
Q8c
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8d vs Q5
Combination
lospital base
niversity base
Percent of responses
0
10
20
30
40
Q8d
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

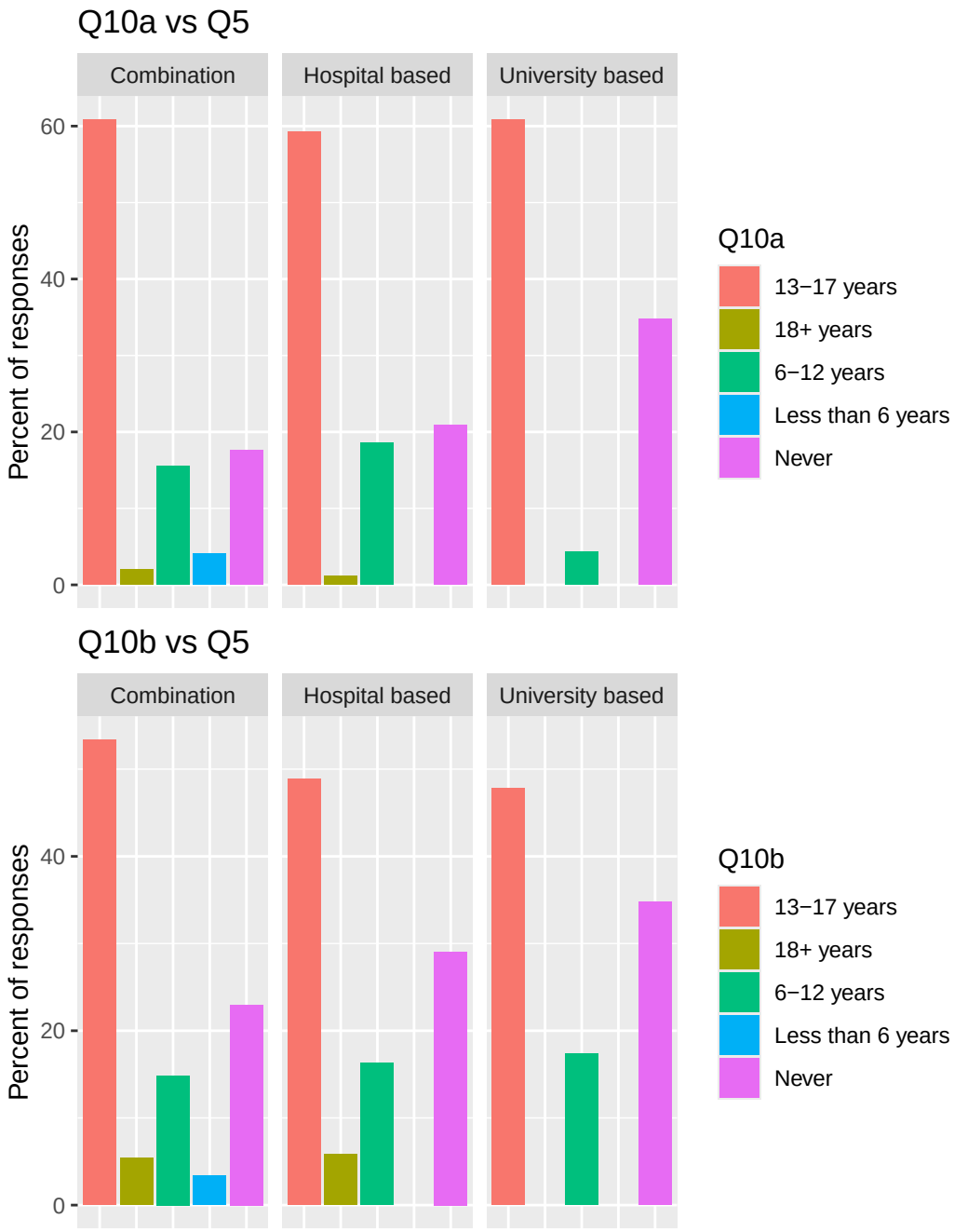

Q8e vs Q5
Combination
lospital base
niversity base
Percent of responses
40
20
0
Q8e
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

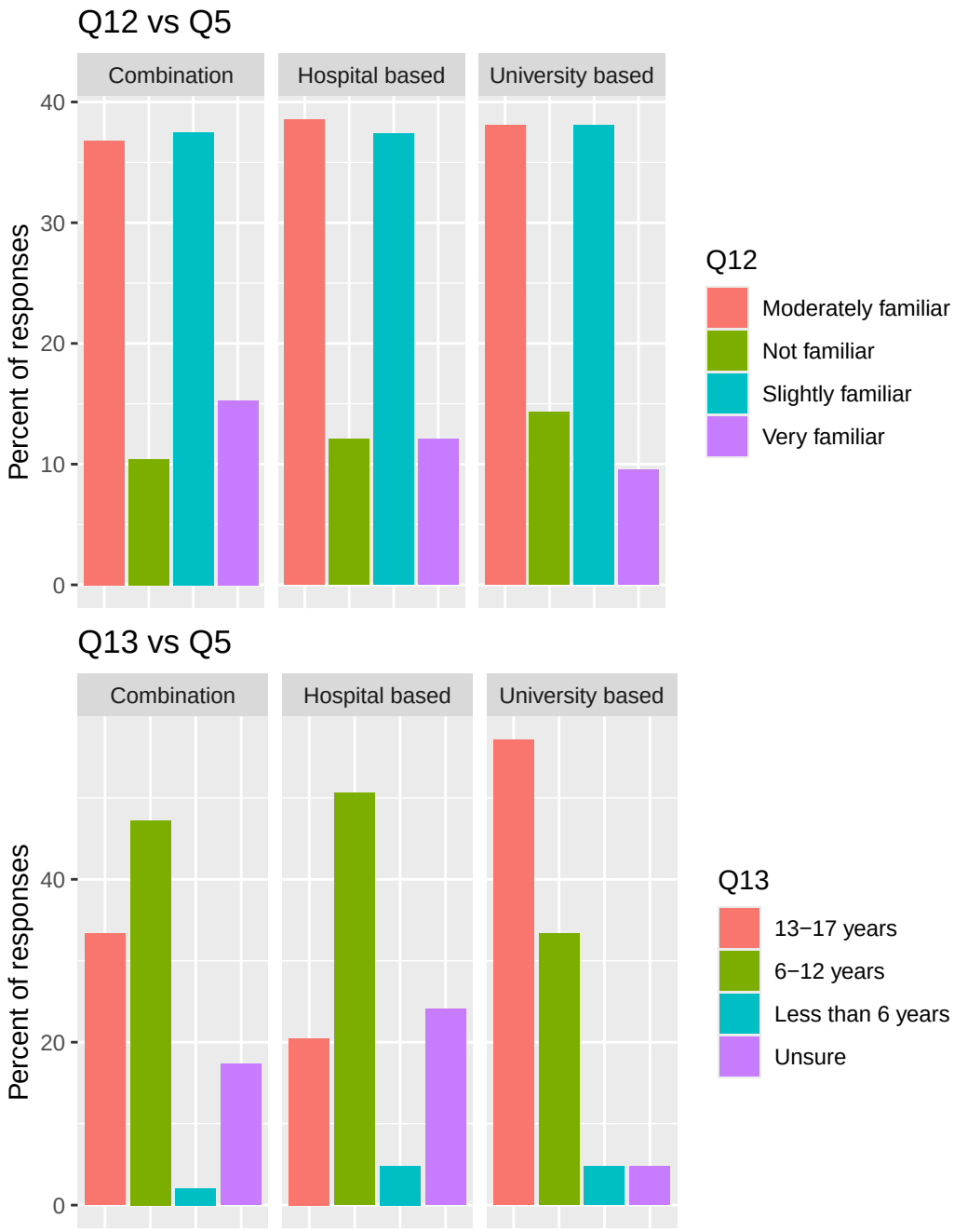

Q8f vs Q5
Combination
lospital base
niversity base
Percent of responses
40
30
20
10
0
Q8f
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

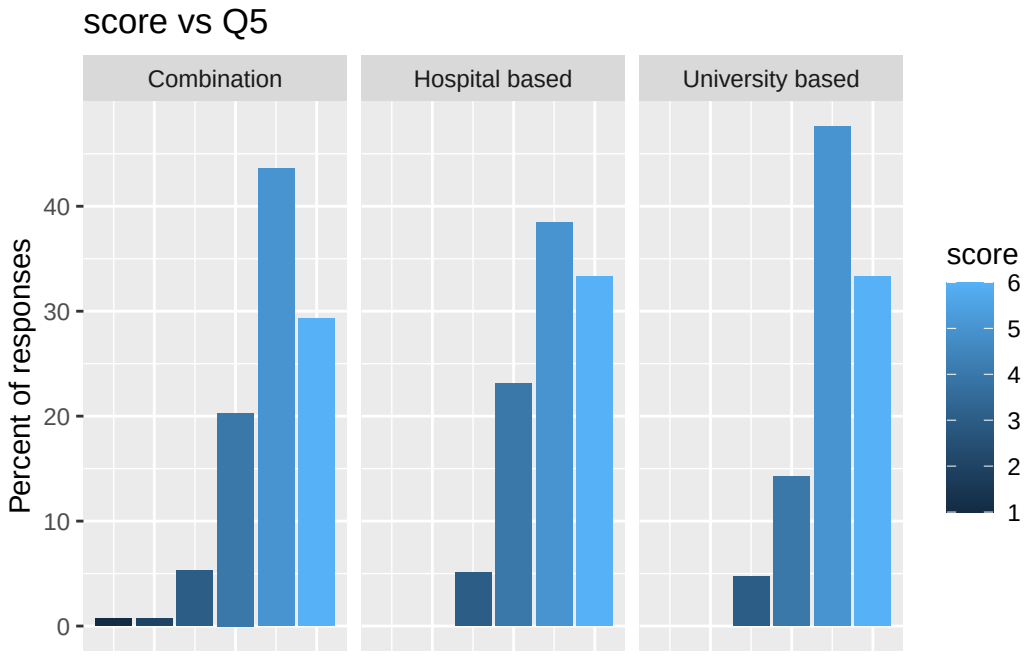

Q8g vs Q5
Combination
lospital base
niversity base
Percent of responses
0
20
40
60
Q8g
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8h vs Q5
Combination
lospital base
niversity base
Percent of responses
0
20
40
60
Q8h
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable


Q8i vs Q5
Combination
lospital base
niversity base
Percent of responses
30
20
10
0
Q8i
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8j vs Q5
Combination
lospital base
niversity base
Percent of responses
30
20
10
0
Q8j
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable


Q8k vs Q5
Combination
Percent of responses
0
10
20
30
40
Q8k
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q9 vs Q5
Combination
Hospital based
University based
Percent of responses
0
10
20
30
40
Q9
Always
Never
Often
Rarely
Sometimes

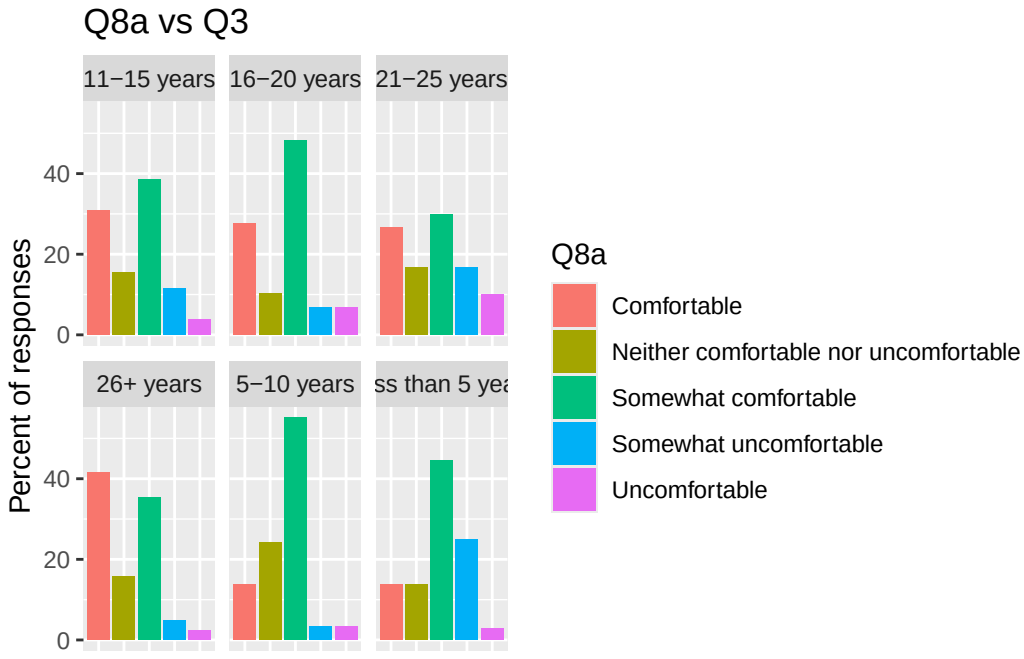

Q10a vs Q5
Combination
Hospital based
University based
Percent of responses
60
40
20
0
Q10a
13−17 years
18+ years
6−12 years
Less than 6 years
Never
Q10b vs Q5
Combination
Hospital based
University based
Percent of responses
40
20
0
Q10b
13−17 years
18+ years
6−12 years
Less than 6 years
Never

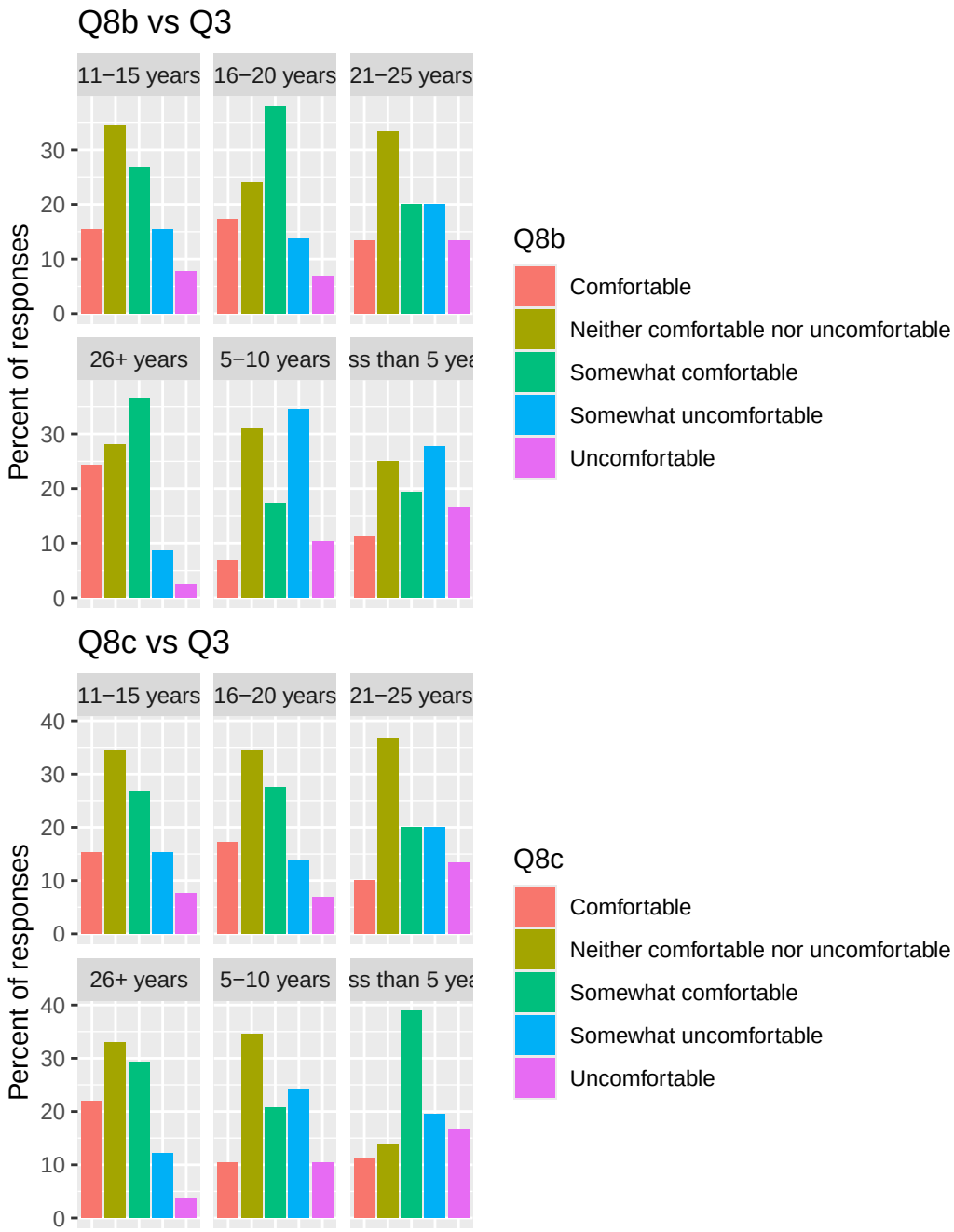

Q12 vs Q5
Combination
Hospital based
University based
Percent of responses
0
10
20
30
40
Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Q13 vs Q5
Combination
Hospital based
University based
Percent of responses
0
20
40
Q13
13−17 years
6−12 years
Less than 6 years
Unsure

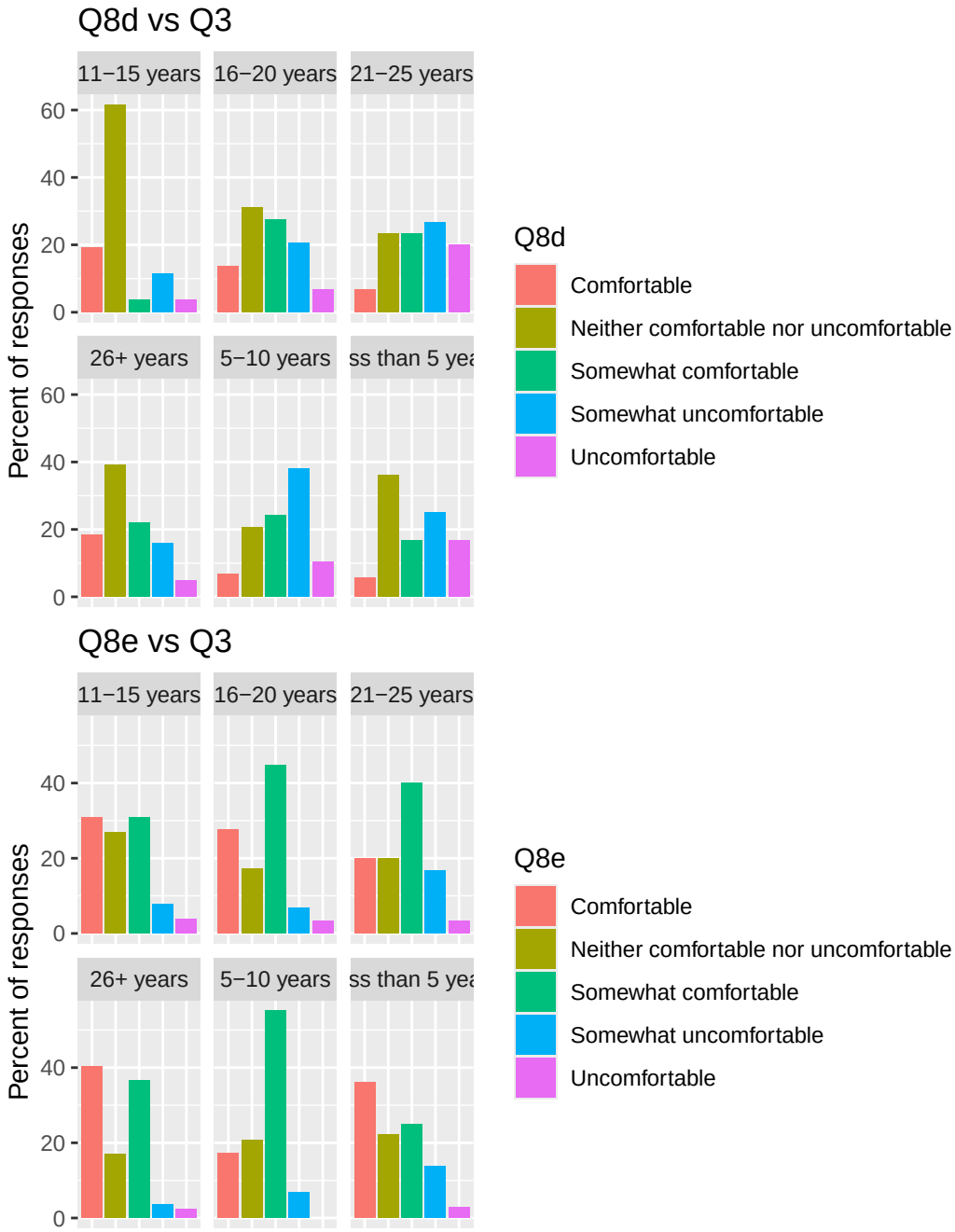


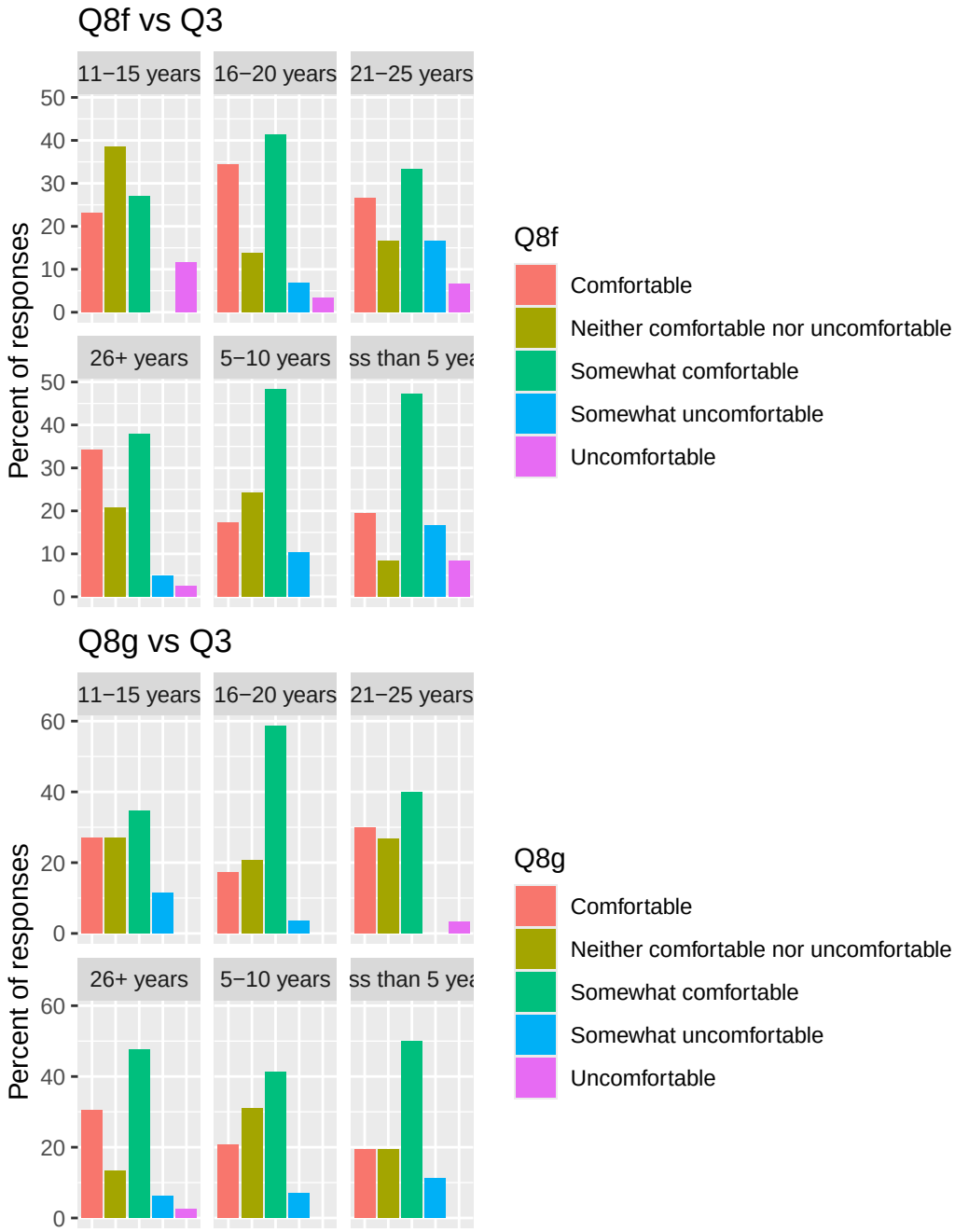
Dentist Analysis Plots

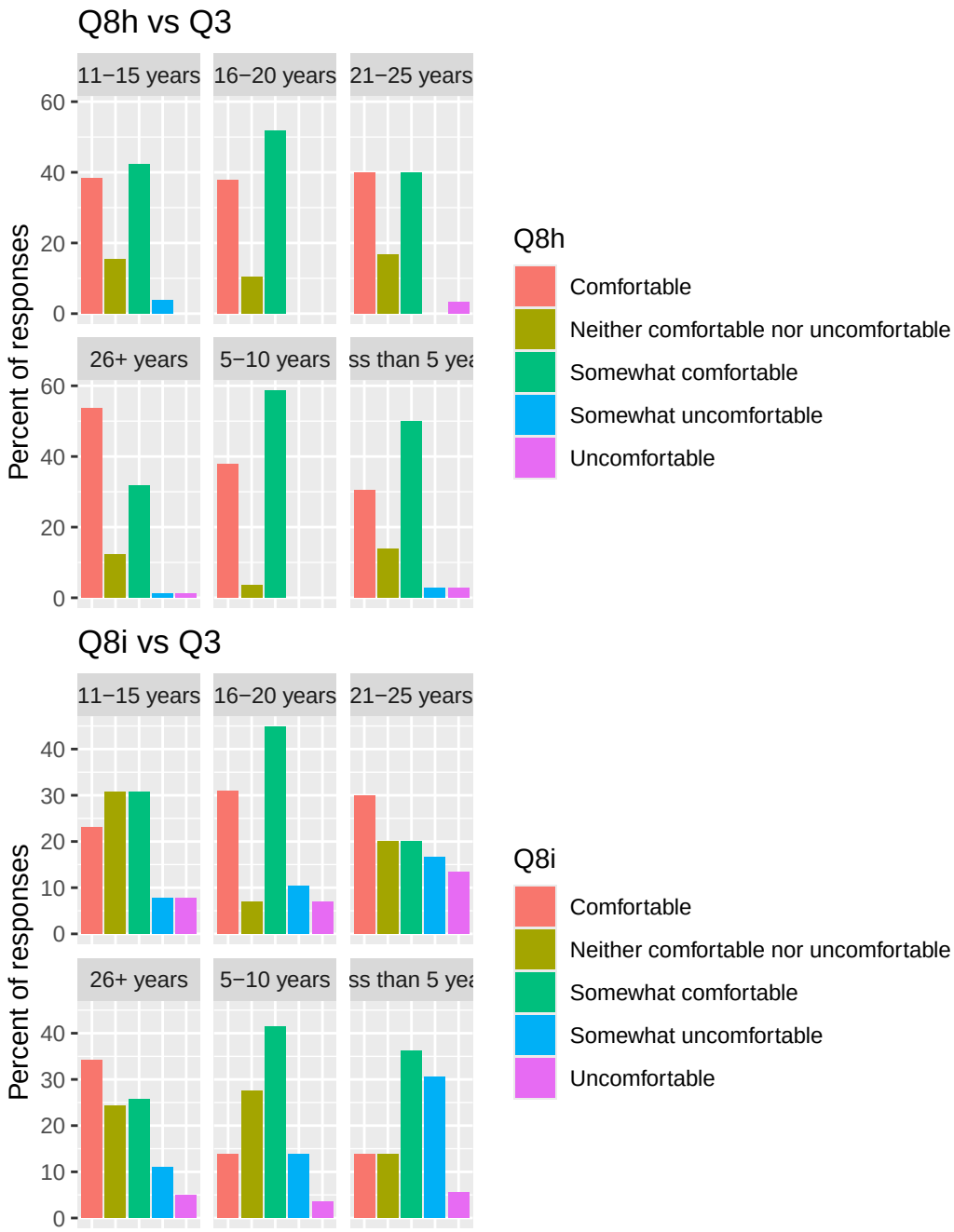
Below are plots for the dentist analysis.

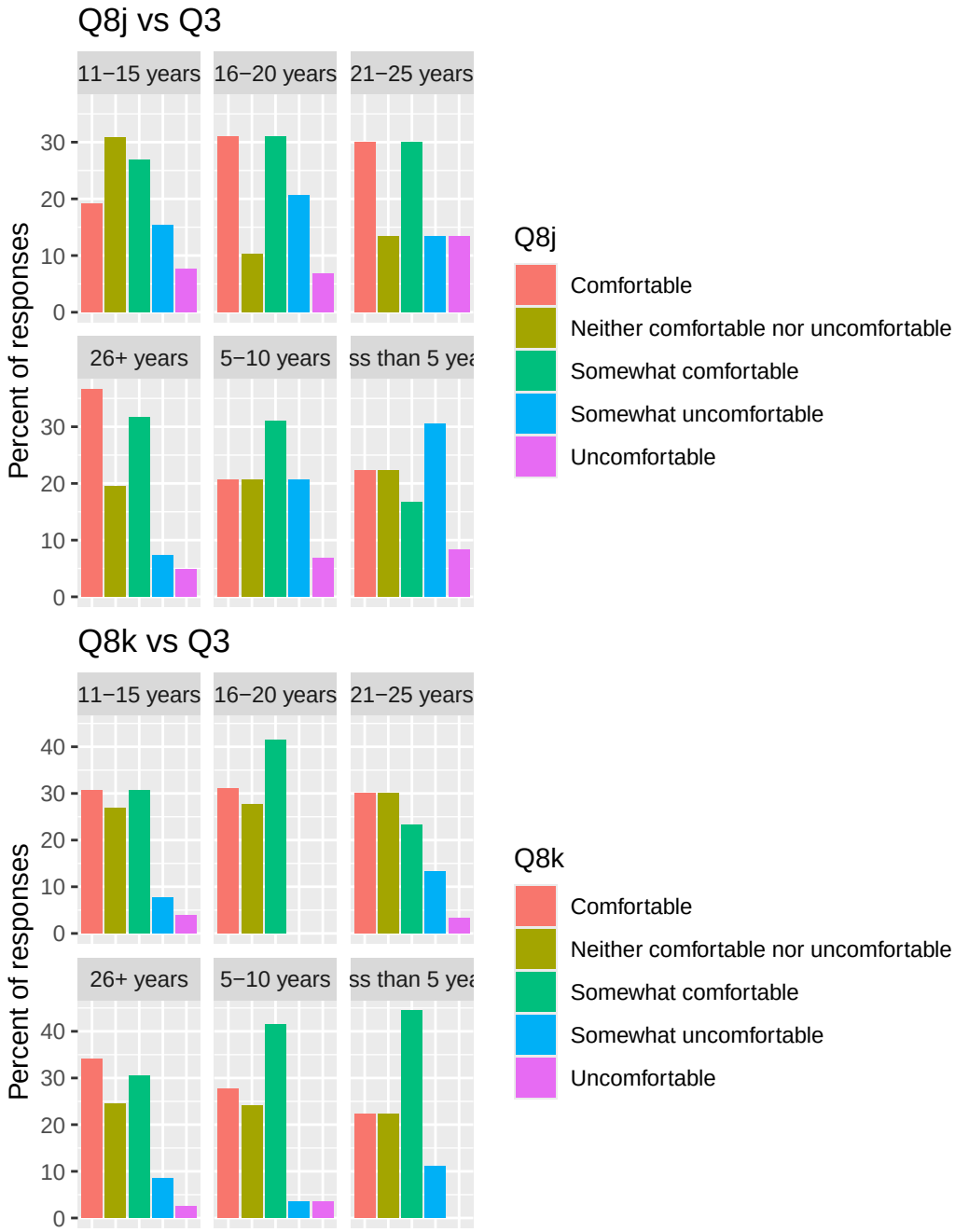



Q8b vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
ss than 5 yea
Percent of responses
0
10
20
30
Q8b
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8c vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
ss than 5 yea
Percent of responses
0
10
20
30
40
Q8c
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

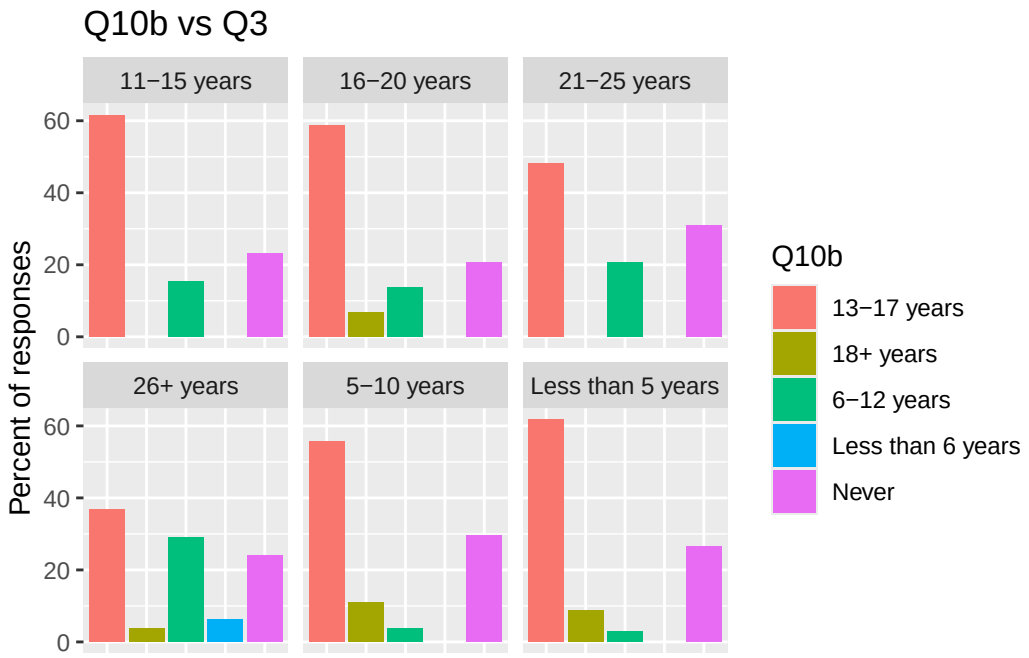

Q8d vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
ss than 5 yea
Percent of responses
0
20
40
60
Q8d
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8e vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
ss than 5 yea
Percent of responses
0
20
40
Q8e
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable


Q8f vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
ss than 5 yea
Percent of responses
0
10
20
30
40
50
Q8f
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8g vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
ss than 5 yea
Percent of responses
0
20
40
60
Q8g
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable


Q8h vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
ss than 5 yea
Percent of responses
0
20
40
60
Q8h
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8i vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
ss than 5 yea
Percent of responses
0
10
20
30
40
Q8i
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

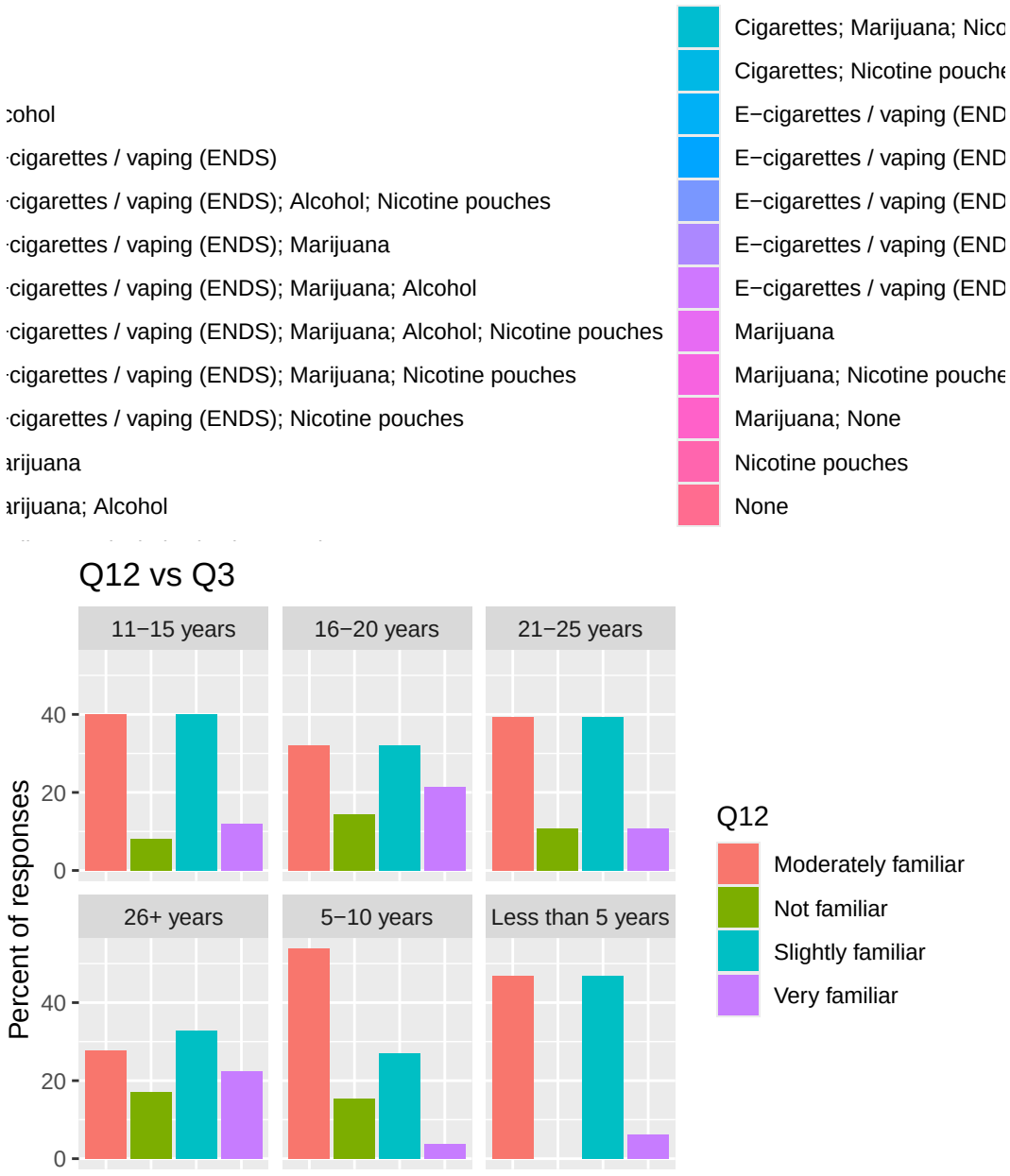

Q8j vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
ss than 5 yea
Percent of responses
0
10
20
30
Q8j
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8k vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
ss than 5 yea
Percent of responses
0
10
20
30
40
Q8k
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

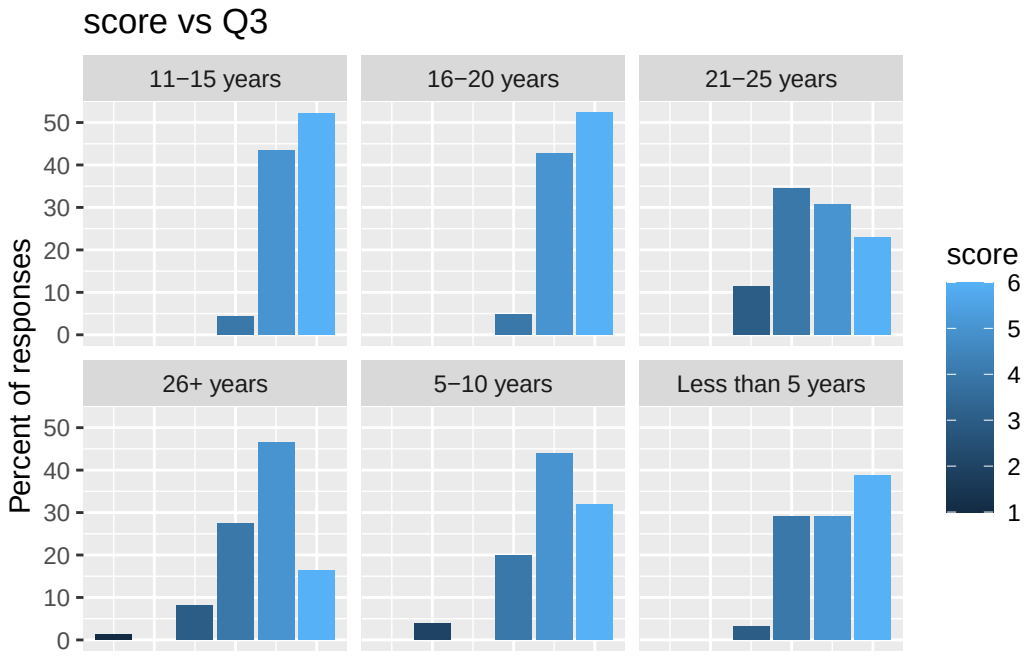

Q9 vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
Less than 5 years
Percent of responses
0
10
20
30
40
Q9
Always
Never
Often
Rarely
Sometimes
Q10a vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
Less than 5 years
Percent of responses
0
20
40
60
Q10a
13−17 years
18+ years
6−12 years
Less than 6 years
Never



es / vaping (ENDS)
es / vaping (ENDS); Alcohol
es / vaping (ENDS); Alcohol; Nicotine pouches
es / vaping (ENDS); Marijuana
es / vaping (ENDS); Marijuana; Alcohol
es / vaping (ENDS); Marijuana; Alcohol; Nicotine pouches
es / vaping (ENDS); Marijuana; Alcohol; Nicotine pouches; None
es / vaping (ENDS); Marijuana; Nicotine pouches
es / vaping (ENDS); Nicotine pouches

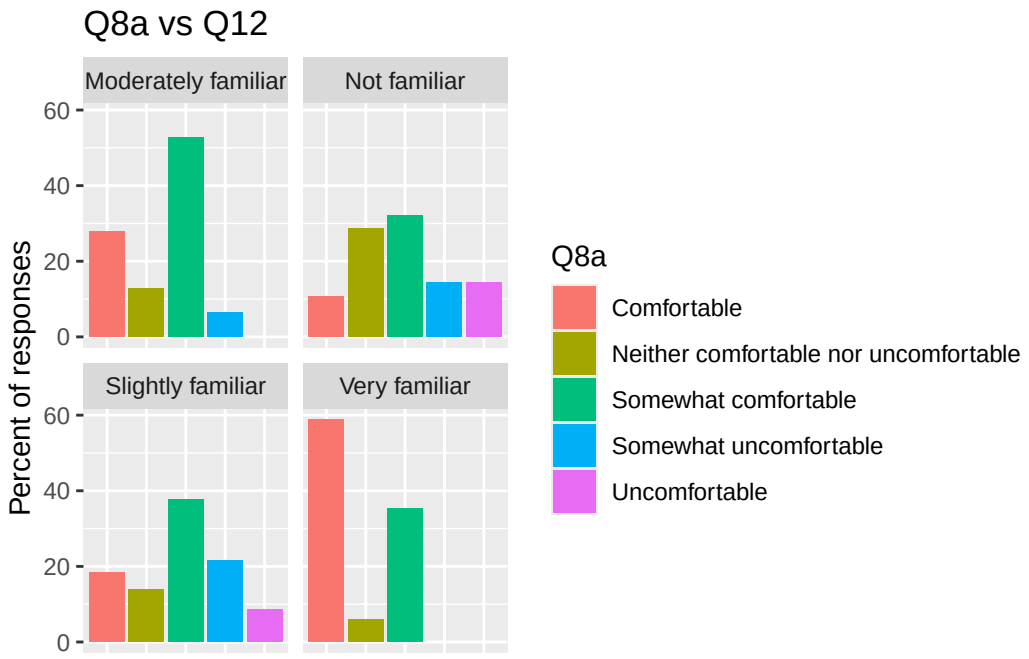
Cigarettes; Marijuana; Alcoh
Cigarettes; Nicotine pouche
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
Marijuana
Marijuana; Alcohol
Nicotine pouches

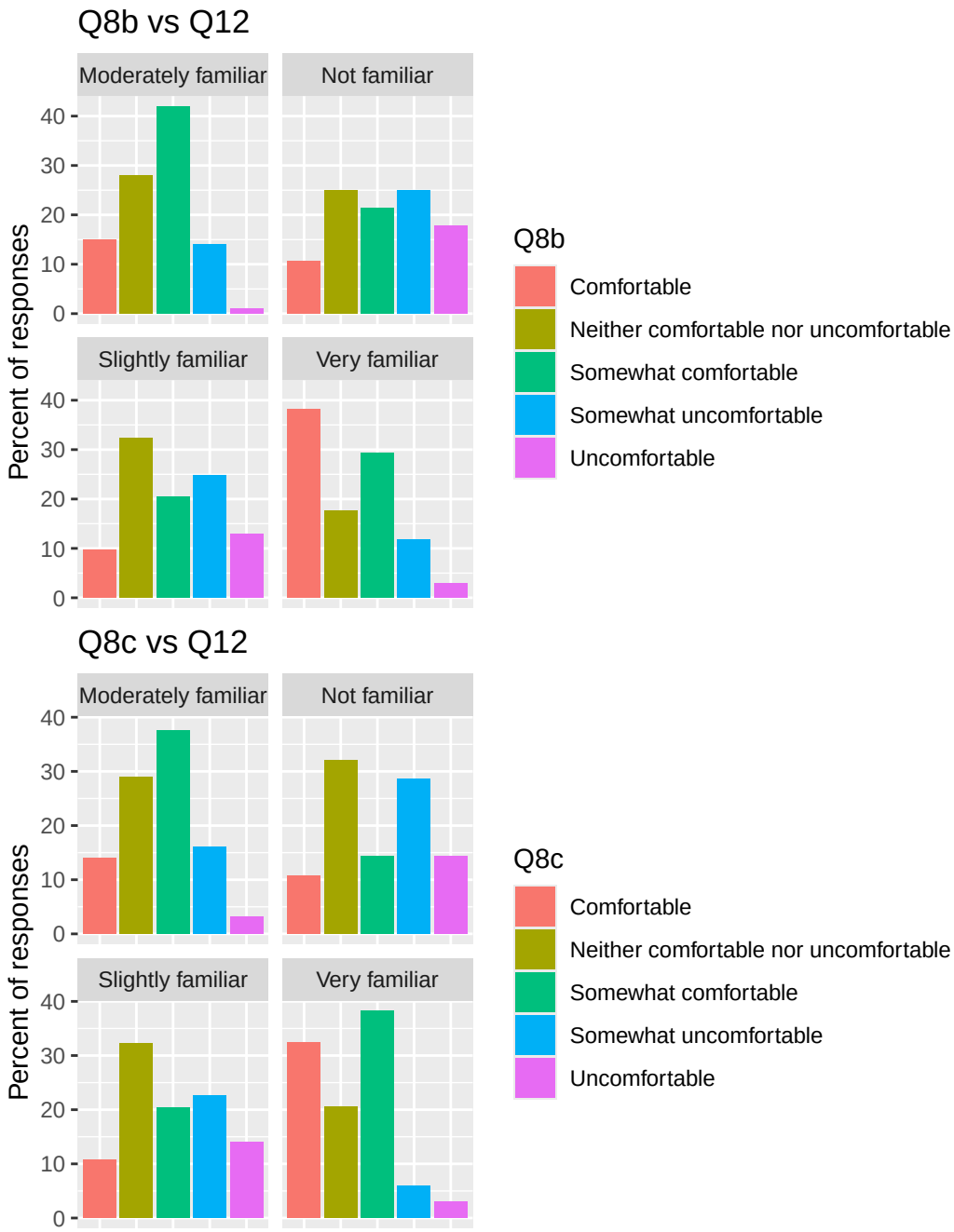

cohol
cigarettes / vaping (ENDS)
cigarettes / vaping (ENDS); Alcohol; Nicotine pouches
cigarettes / vaping (ENDS); Marijuana
cigarettes / vaping (ENDS); Marijuana; Alcohol
cigarettes / vaping (ENDS); Marijuana; Alcohol; Nicotine pouches
cigarettes / vaping (ENDS); Marijuana; Nicotine pouches
cigarettes / vaping (ENDS); Nicotine pouches
arijuana
arijuana; Alcohol
Cigarettes; Marijuana; Nicc
Cigarettes; Nicotine pouche
E−cigarettes / vaping (END
E−cigarettes / vaping (END
E−cigarettes / vaping (END
E−cigarettes / vaping (END
E−cigarettes / vaping (END
Marijuana
Marijuana; Nicotine pouche
Marijuana; None
Nicotine pouches
None
Q12 vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
Less than 5 years
Percent of responses
40
20
0
Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar

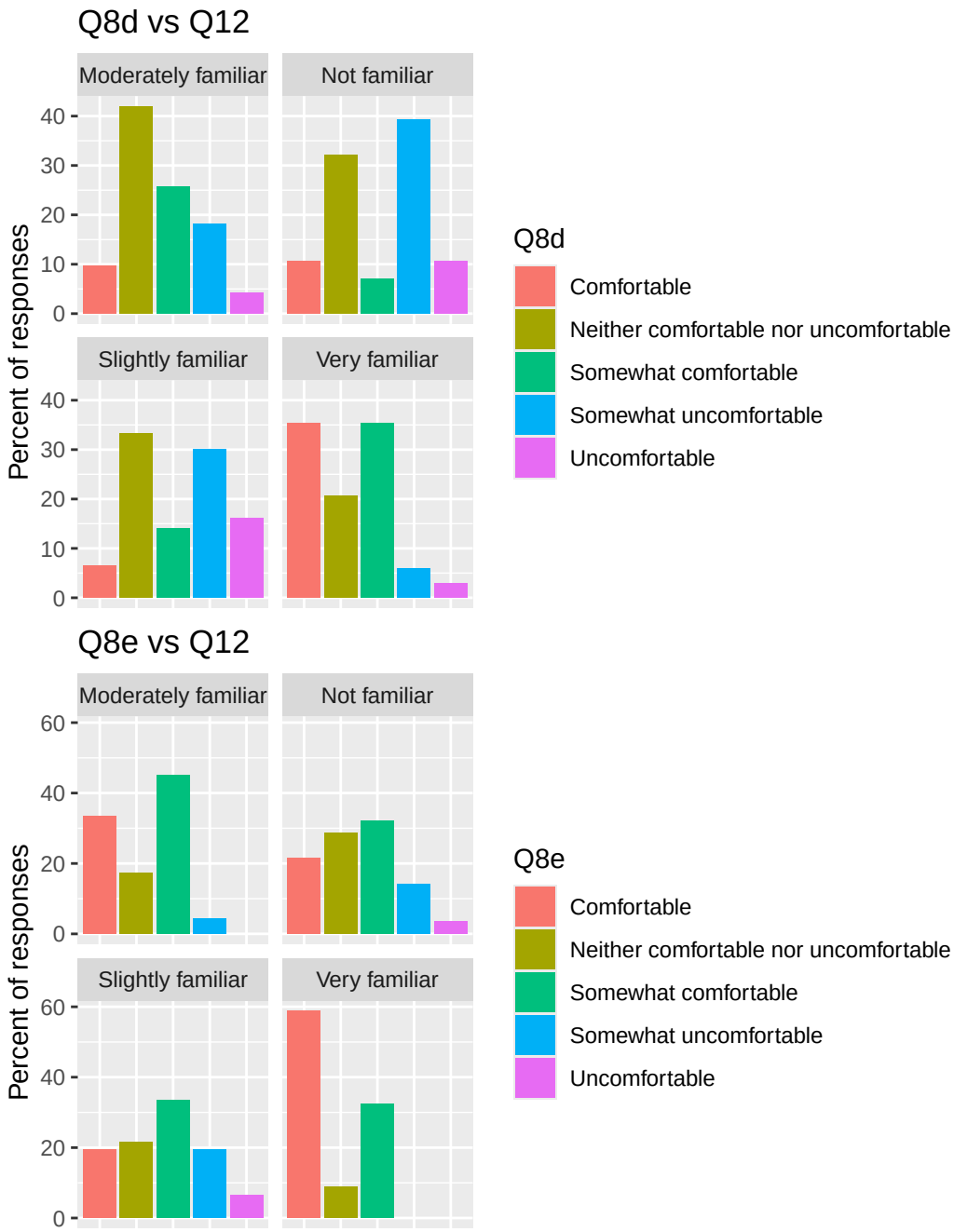


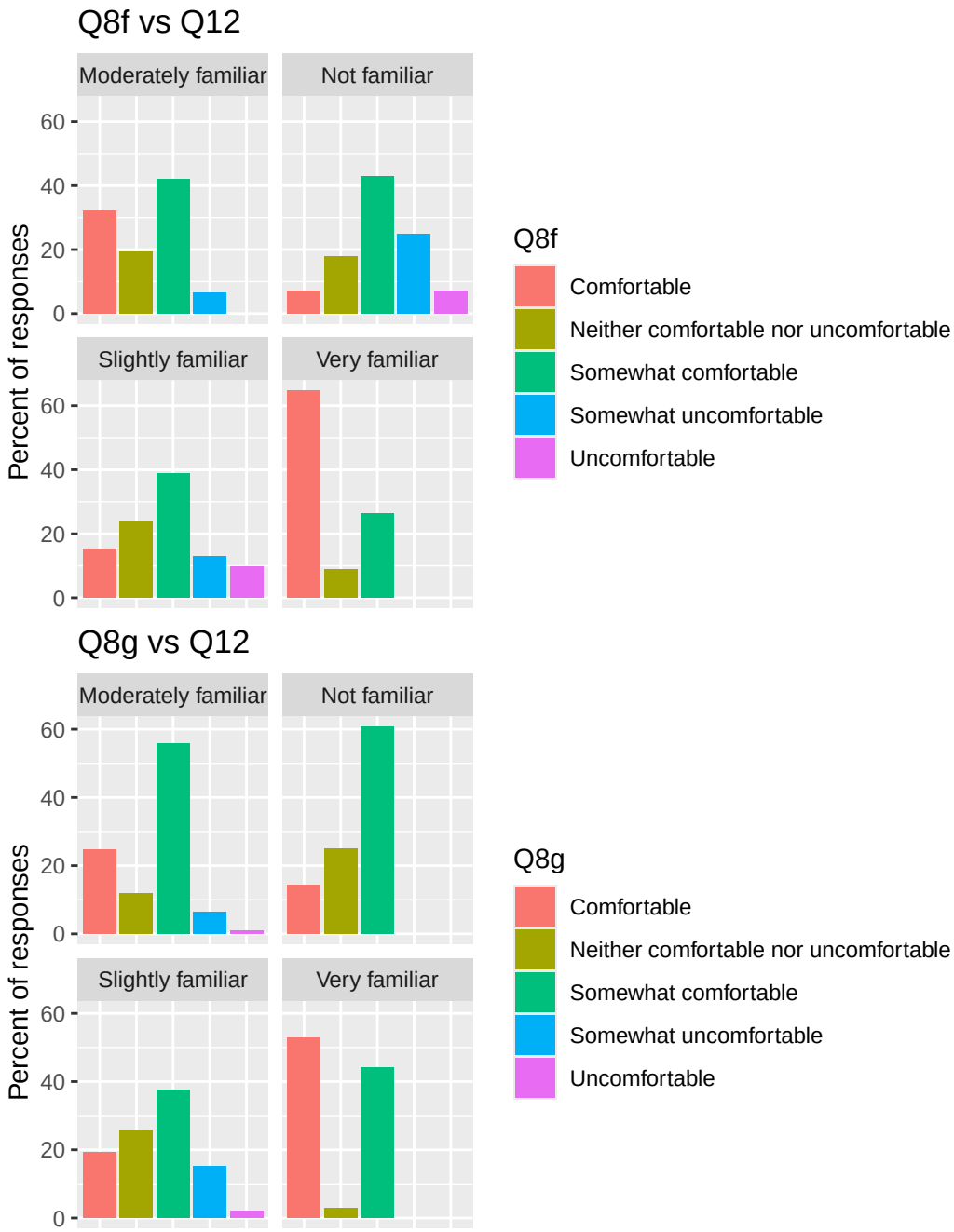
Additional Analysis Plots

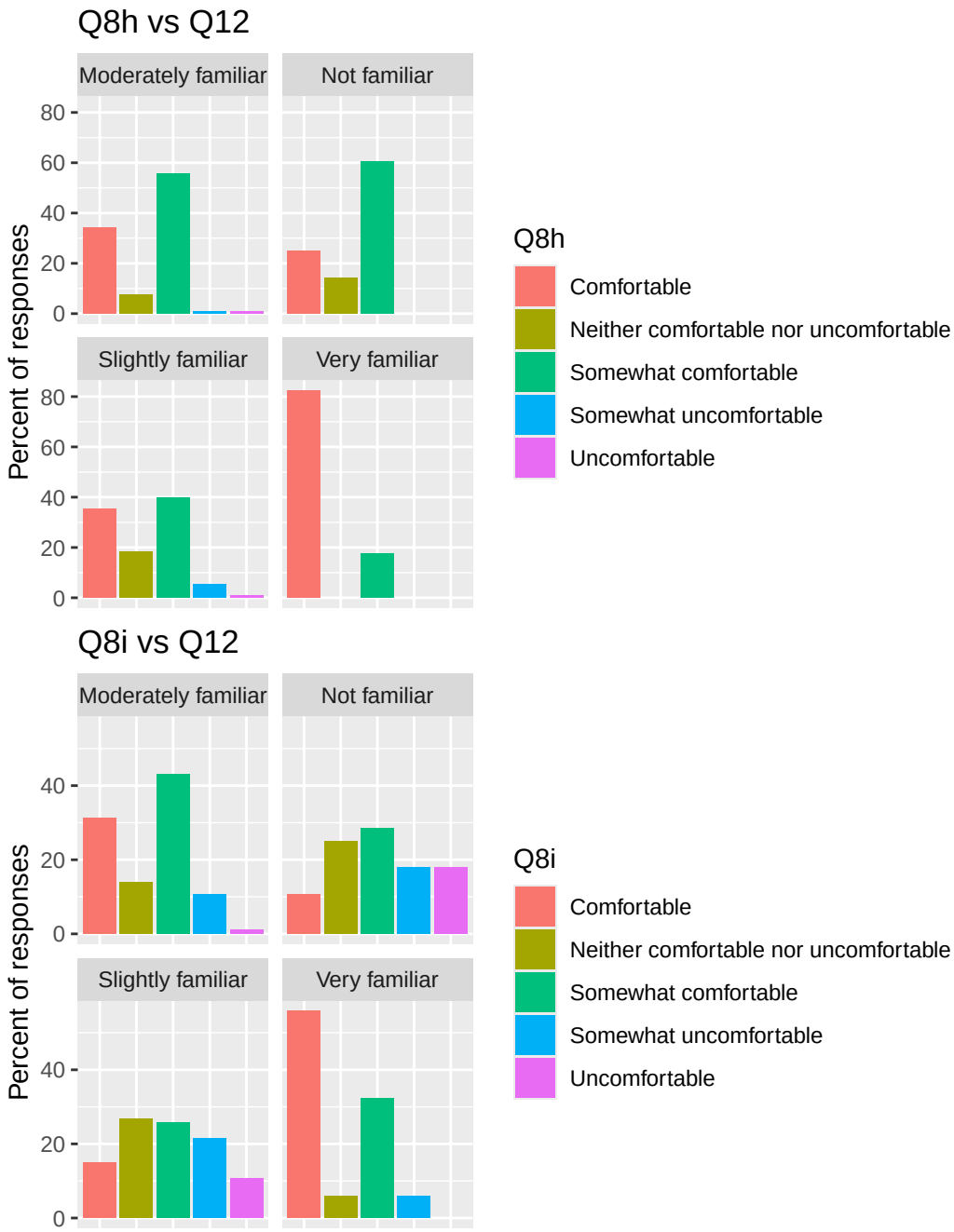
Below are plots comparing each additional analysis question.

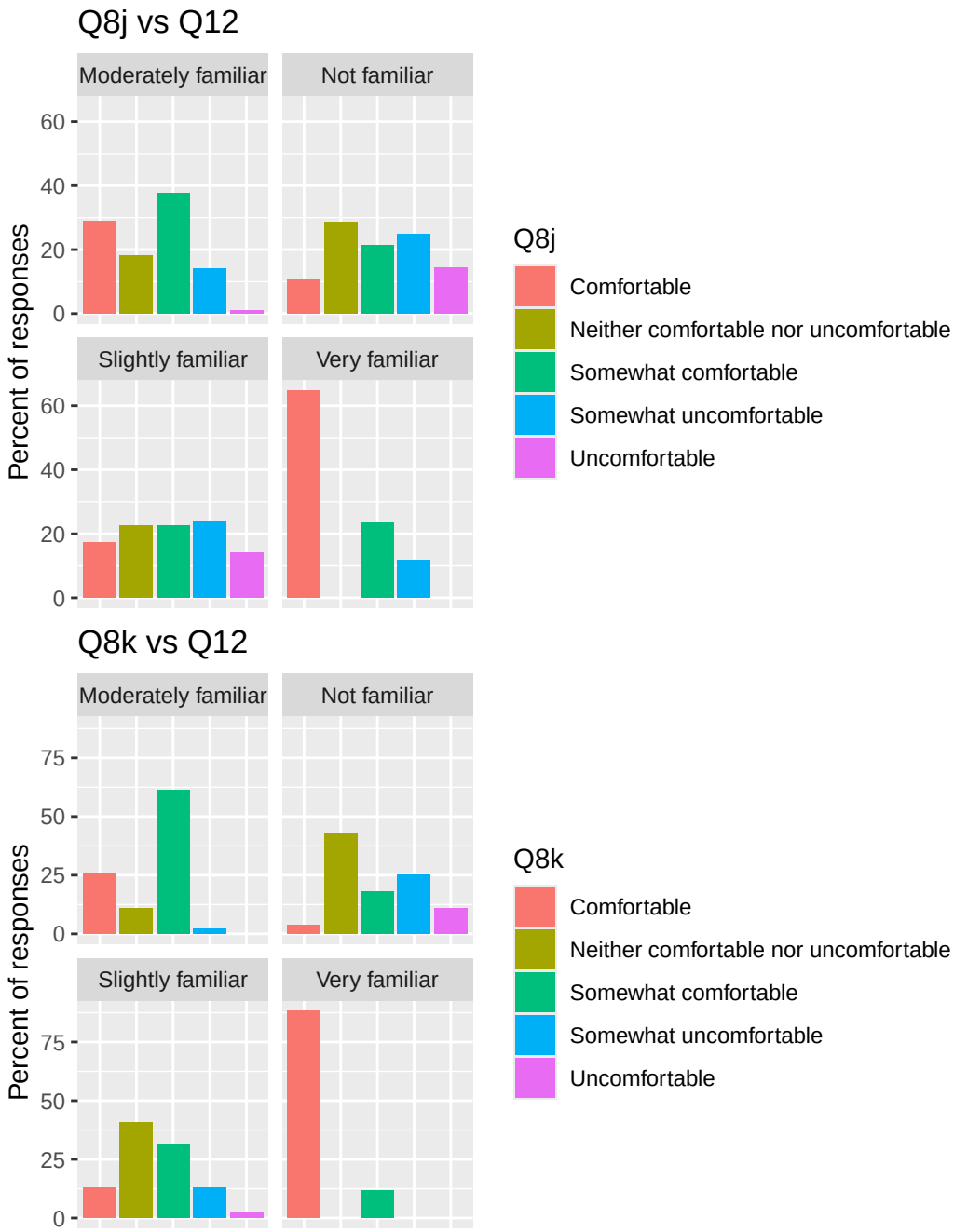


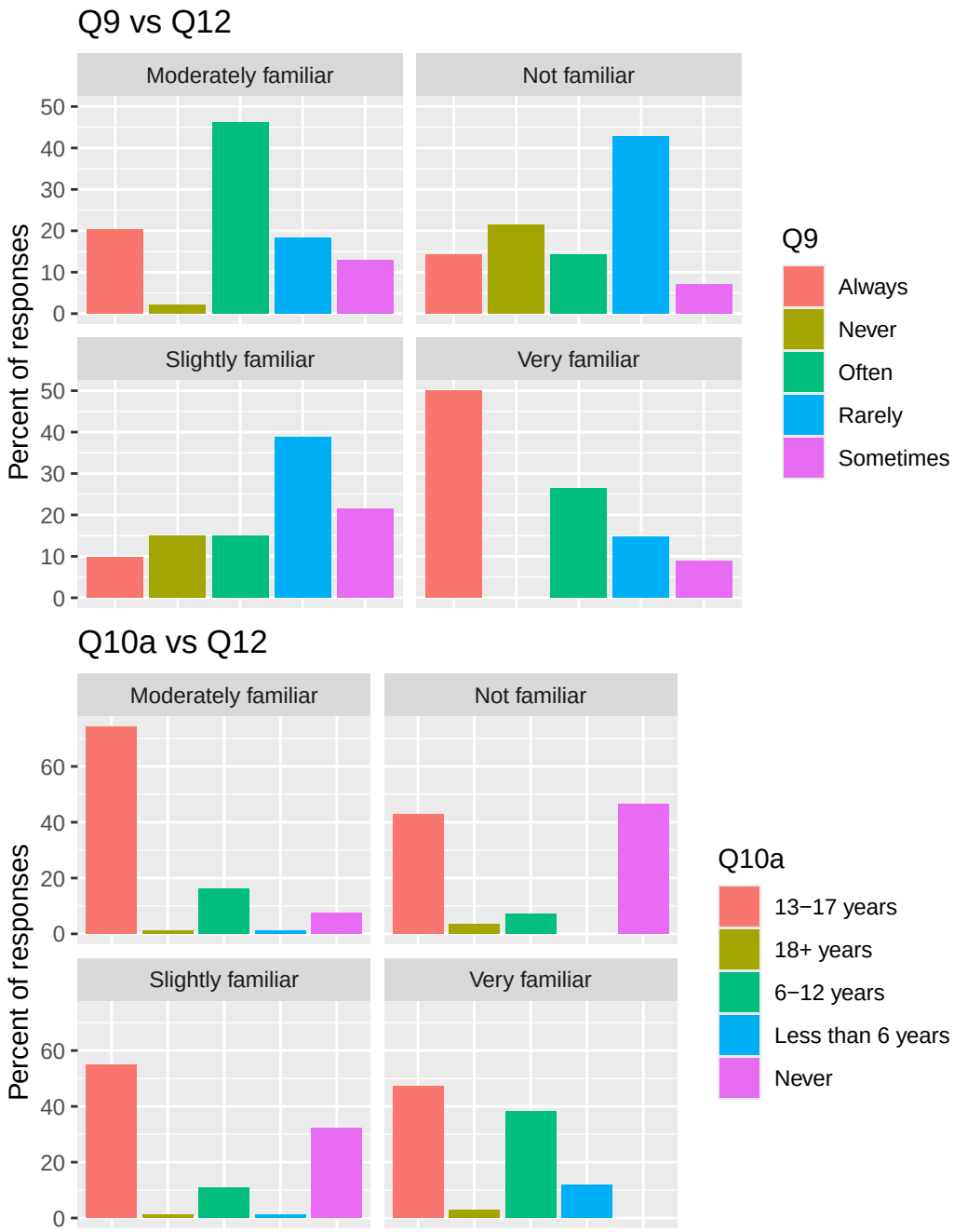

Q8b vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
10
20
30
40
Q8b
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8c vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
10
20
30
40
Q8c
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

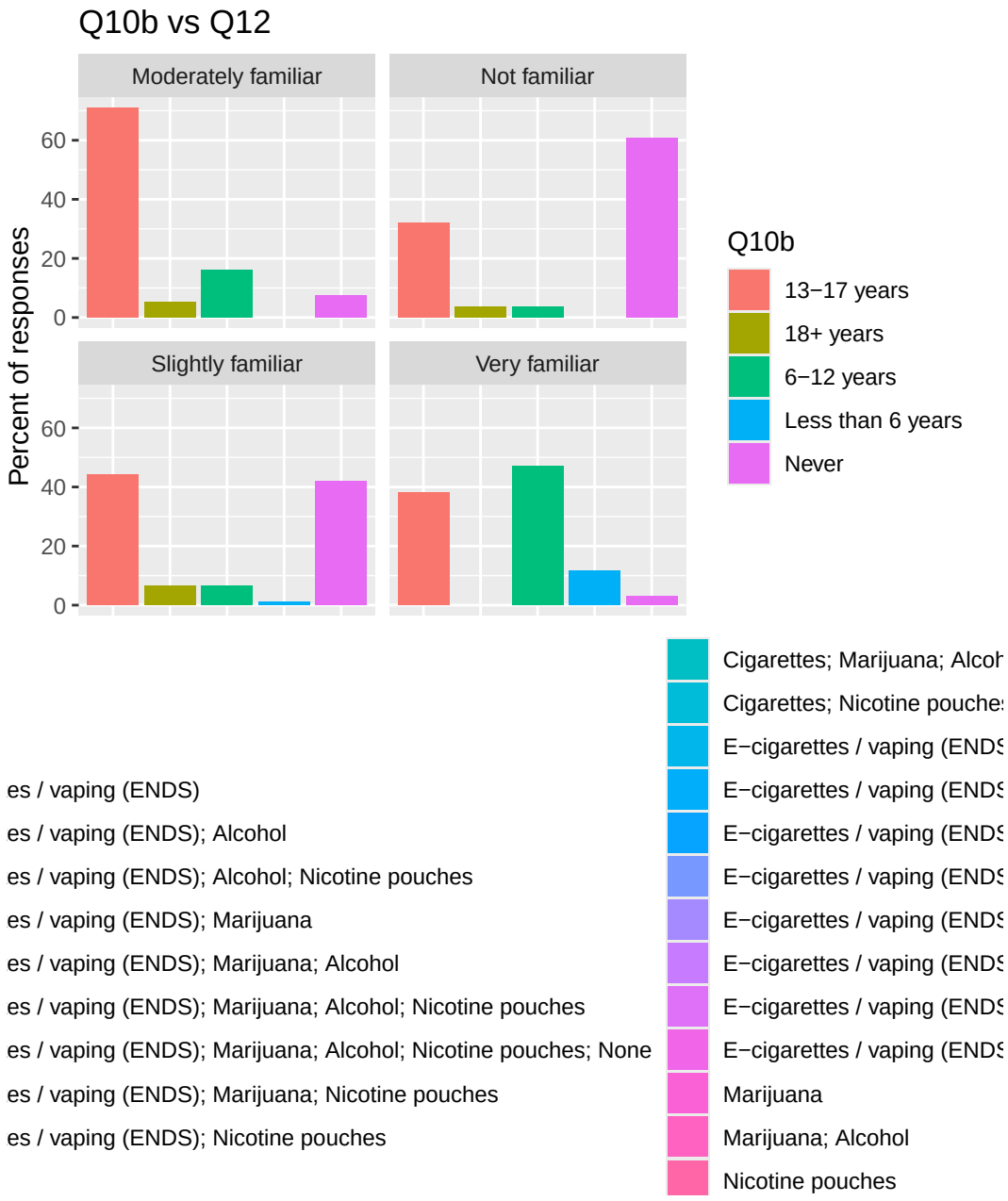

Q8d vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
10
20
30
40
Q8d
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8e vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
20
40
60
Q8e
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

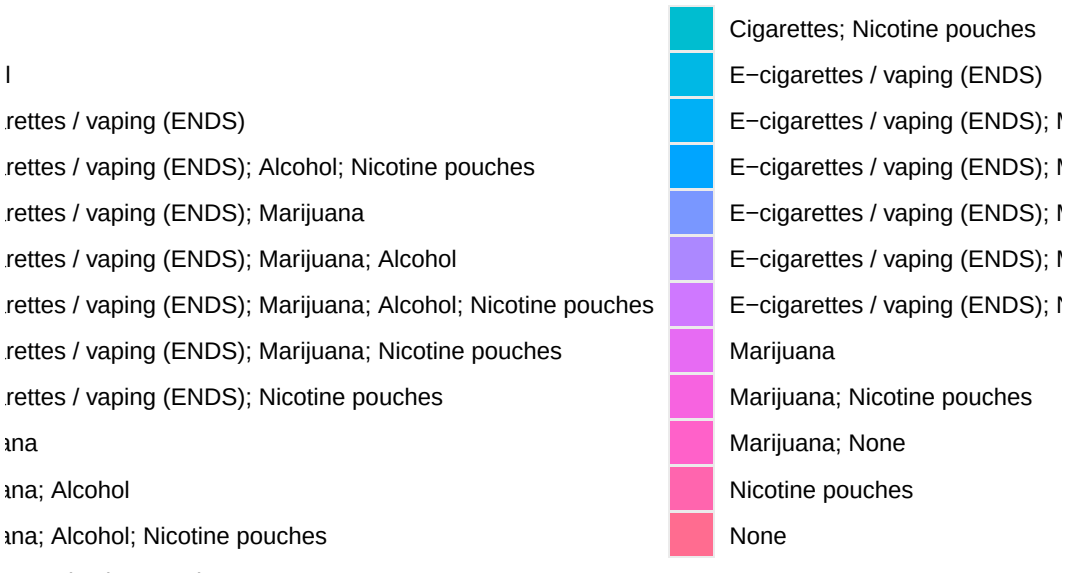

Q8f vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
20
40
60
Q8f
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8g vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
20
40
60
Q8g
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

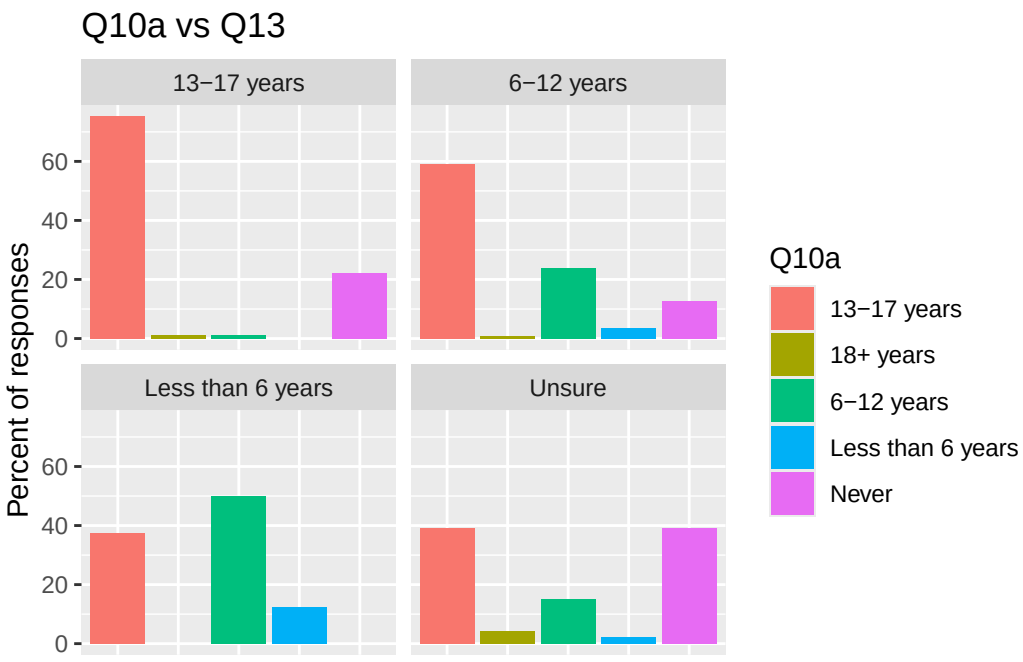

Q8h vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
20
40
60
80
Q8h
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8i vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
20
40
Q8i
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

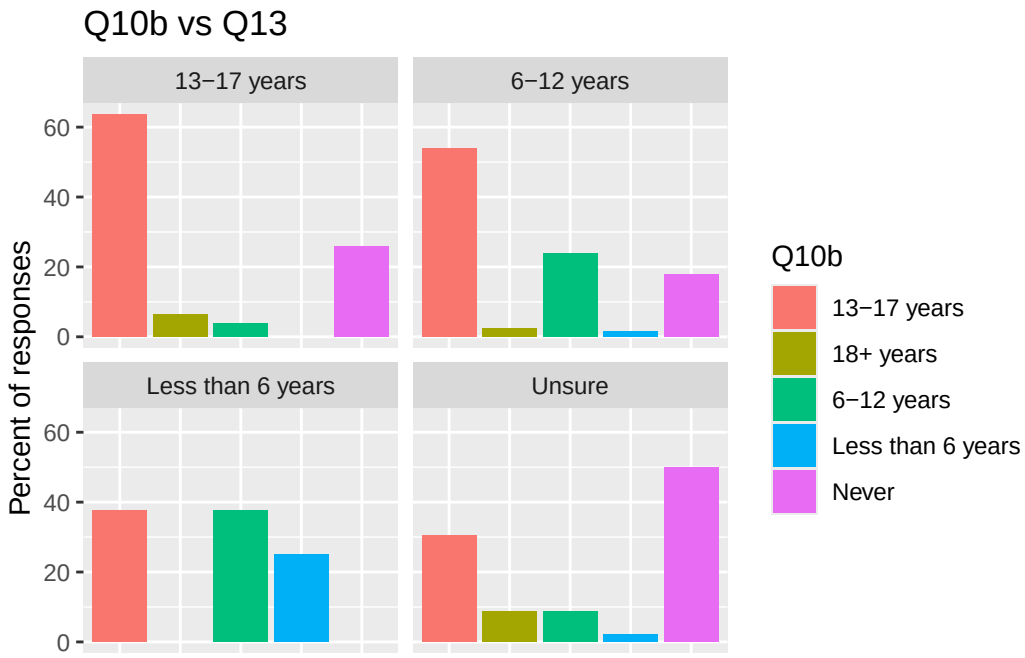

Q8j vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
20
40
60
Q8j
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8k vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
25
50
75
Q8k
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable


Q9 vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
10
20
30
40
50
Q9
Always
Never
Often
Rarely
Sometimes
Q10a vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
20
40
60
Q10a
13−17 years
18+ years
6−12 years
Less than 6 years
Never


Q10b vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
20
40
60
Q10b
13−17 years
18+ years
6−12 years
Less than 6 years
Never
es / vaping (ENDS)
es / vaping (ENDS); Alcohol
es / vaping (ENDS); Alcohol; Nicotine pouches
es / vaping (ENDS); Marijuana
es / vaping (ENDS); Marijuana; Alcohol
es / vaping (ENDS); Marijuana; Alcohol; Nicotine pouches
es / vaping (ENDS); Marijuana; Alcohol; Nicotine pouches; None
es / vaping (ENDS); Marijuana; Nicotine pouches
es / vaping (ENDS); Nicotine pouches
Cigarettes; Marijuana; Alcoh
Cigarettes; Nicotine pouche
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
Marijuana
Marijuana; Alcohol
Nicotine pouches


l
rettes / vaping (ENDS)
rettes / vaping (ENDS); Alcohol; Nicotine pouches
rettes / vaping (ENDS); Marijuana
rettes / vaping (ENDS); Marijuana; Alcohol
rettes / vaping (ENDS); Marijuana; Alcohol; Nicotine pouches
rettes / vaping (ENDS); Marijuana; Nicotine pouches
rettes / vaping (ENDS); Nicotine pouches
ana
ana; Alcohol
ana; Alcohol; Nicotine pouches
Cigarettes; Nicotine pouches
E−cigarettes / vaping (ENDS)
E−cigarettes / vaping (ENDS); M
E−cigarettes / vaping (ENDS); M
E−cigarettes / vaping (ENDS); M
E−cigarettes / vaping (ENDS); M
Marijuana
Marijuana; Nicotine pouches
Marijuana; None
Nicotine pouches
None


Q10a vs Q13
13−17 years
6−12 years
Less than 6 years
Unsure
Percent of responses
60
40
20
0
Q10a
13−17 years
18+ years
6−12 years
Less than 6 years
Never



Appendix B: Code

```
library(tidyverse)
library(zoo)
library(stringr)
library(gt)
set.seed(1)
m_headers <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/McGannData.csv", n_max = 2, col

questions <- as.character(m_headers[1,])

answer_choices <- as.character(m_headers[2,])

questions_filled <- na.locf(ifelse(questions == "" | is.na(questions), NA, questions))

clean_headers <- paste(questions_filled, answer_choices, sep = "_")

m_rawdata <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/McGannData.csv", skip = 2, col_

m_non_coalesced <- m_rawdata %>%
  select(c(10:ncol(m_rawdata)))
```

```r
m_non_coalesced <- m_non_coalesced %>%
  mutate(
    across(
      everything(),
      ~ na_if(as.character(.), "")
    )
  )

m_questions <- m_non_coalesced %>%
  mutate(Q1 = do.call(coalesce, pick(1:6))) %>%
  mutate(Q2 = do.call(coalesce, pick(7:12))) %>%
  mutate(Q3 = do.call(coalesce, pick(13:21))) %>%
  mutate(Q4 = do.call(coalesce, pick(22:24))) %>%
  mutate(Q5 = do.call(coalesce, pick(25:27))) %>%
  mutate(Q6 = do.call(coalesce, pick(28:33))) %>%
  mutate(Q7 = do.call(coalesce, pick(34:40))) %>%
  mutate(Q8a = do.call(coalesce, pick(41:45))) %>%
  mutate(Q8b = do.call(coalesce, pick(46:50))) %>%
  mutate(Q8c = do.call(coalesce, pick(51:55))) %>%
  mutate(Q8d = do.call(coalesce, pick(56:60))) %>%
  mutate(Q8e = do.call(coalesce, pick(61:65))) %>%
  mutate(Q8f = do.call(coalesce, pick(66:70))) %>%
  mutate(Q8g = do.call(coalesce, pick(71:75))) %>%
  mutate(Q8h = do.call(coalesce, pick(76:80))) %>%
  mutate(Q8i = do.call(coalesce, pick(81:85))) %>%
  mutate(Q8j = do.call(coalesce, pick(86:90))) %>%
  mutate(Q8k = do.call(coalesce, pick(91:95))) %>%
  mutate(Q9 = do.call(coalesce, pick(96:100))) %>%
  mutate(Q10a = do.call(coalesce, pick(101:105))) %>%
  mutate(Q10b = do.call(coalesce, pick(106:110))) %>%
  #mutate(Q11a = do.call(coalesce, pick(111:116))) %>%
  #mutate(Q11b = do.call(coalesce, pick(117:122))) %>%
  mutate(Q12 = do.call(coalesce, pick(123:126))) %>%
  mutate(Q13 = do.call(coalesce, pick(127:131))) %>%
  mutate(Q14 = do.call(coalesce, pick(132:137))) %>%
  mutate(Q15 = do.call(coalesce, pick(138:141))) %>%
  mutate(Q16 = do.call(coalesce, pick(142:145))) %>%
  mutate(Q17 = do.call(coalesce, pick(146:149))) %>%
  mutate(Q18 = do.call(coalesce, pick(150:153))) %>%
  mutate(Q19 = do.call(coalesce, pick(154:157))) %>%
  mutate(Q20 = do.call(coalesce, pick(158:161)))
  #mutate(Q21 = do.call(coalesce, pick(162:169))) %>%
```

```
  #mutate(Q22 = do.call(coalesce, pick(170:175))) %>%
  #mutate(Q23 = do.call(coalesce, pick(176:183))) %>%

m_no_sata <- m_questions %>%
  select(184:204, 111:122, 205:213, 162:183)

m_coalesced <- m_no_sata %>%
  mutate(Q11a = pmap_chr(select(., 22:27), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  mutate(Q11b = pmap_chr(select(., 28:33), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  mutate(Q21 = pmap_chr(select(., 45:50), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  mutate(Q22 = pmap_chr(select(., 51:56), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  mutate(Q23 = pmap_chr(select(., 57:64), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  select(1:21, Q11a, Q11b, 34:42, Q21, Q22, Q23)

coll_ids <- m_rawdata$`Collector ID_NA`

m_coalesced <- m_coalesced %>%
  mutate(ids = coll_ids) %>%
  mutate(role = case_when(
    ids == 440127144 ~ "Dentist",
    ids == 440133230 ~ "Resident"
  ))

correct_answers <- c("Ask about tobacco or vaping use and provide counseling", "Counsel on to

m_coalesced <- m_coalesced %>%
  mutate(
    score = if_else(
      if_all(Q15:Q20, is.na),
      NA_real_,
      rowSums(
        across(Q15:Q20) == matrix(
          correct_answers,
          nrow = n(),
          ncol = length(correct_answers),
          byrow = TRUE
        ),
        na.rm = TRUE
      )
    )
  )
validation <- function(question1, question2, data){
```

```r
  df <- data %>%
    transmute(
      response1 = .data[[question1]],
      response2 = .data[[question2]]
    ) %>%
    drop_na()

  comparison_res <- df %>%
    group_by(response1, response2) %>%
    summarize(counts = n(), .groups = "drop") %>%
    group_by(response2) %>%
    mutate(percent = counts / sum(counts) * 100) %>%
    ungroup()

  table_comparison <- table(df$response2, df$response1)
  #table_hours
  test <- suppressWarnings(chisq.test(table_comparison))

  if (mean(test$expected <= 5) < 0.8 | any(test$expected < 1)){
    fish_test <- fisher.test(table_comparison, simulate.p.value = TRUE)
    if (fish_test$p.value <= 0.05){
      significant <- TRUE
    }
    else {
      significant <- FALSE
    }
    return(c("question1", "question2", round(fish_test$p.value, digits = 4), significant, "fi
  }
  else{
    if (test$p.value <= 0.05){
      significant <- TRUE
    }
    else {
      significant <- FALSE
    }
  }
  return(c("question1", "question2", round(test$p.value, digits = 4), significant, "chisq"))
}
comfort_dic <- data.frame(
  q_ref = c("Q8a",  "Q8b",  "Q8c",  "Q8d",  "Q8e",  "Q8f", "Q8g",  "Q8h" , "Q8i" , "Q8j" , "(
  role = c(rep("role", 16))
)
```

```r
comfort_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
  Significant = logical(),
  Test = character()
)

for (i in seq_len(nrow(comfort_dic))) {

  comfort_results[i, ] <- validation(
    question1 = comfort_dic$q_ref[i],
    question2 = comfort_dic$role[i],
    data = m_coalesced
  )
}

comfort_results <- comfort_results %>%
  mutate(
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
    Sig_Adj = Adj_P <= 0.05,
    Question1 = comfort_dic$q_ref,
    Question2 = comfort_dic$role
  ) %>%
  arrange(Adj_P)

comfort_results
knowledge_dic <- data.frame(
  q_ref = c("Q12", "Q13", "Q14"),
  role = c(rep("role", 3))
)

knowledge_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
  Significant = logical(),
  Test = character()
)

for (i in seq_len(nrow(knowledge_dic))) {
```

```
  knowledge_results[i, ] <- validation(
    question1 = knowledge_dic$q_ref[i],
    question2 = knowledge_dic$role[i],
    data = m_coalesced
  )
}

knowledge_results <- knowledge_results %>%
  mutate(
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
    Sig_Adj = Adj_P <= 0.05,
    Question1 = knowledge_dic$q_ref,
    Question2 = knowledge_dic$role
  ) %>%
  arrange(Adj_P)

knowledge_results
quiz_dic <- data.frame(
  q_ref = c("Q15", "Q16", "Q17", "Q18", "Q19", "Q20", "score"),
  role = c(rep("role", 7))
)

quiz_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
  Significant = logical(),
  Test = character()
)

for (i in seq_len(nrow(quiz_dic))) {

  quiz_results[i, ] <- validation(
    question1 = quiz_dic$q_ref[i],
    question2 = quiz_dic$role[i],
    data = m_coalesced
  )
}

quiz_results <- quiz_results %>%
  mutate(
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
```

```
    Sig_Adj = Adj_P <= 0.05,
    Question1 = quiz_dic$q_ref,
    Question2 = quiz_dic$role
  ) %>%
  arrange(Adj_P)

quiz_results
env_dic <- data.frame(
  q_ref = c("Q8a",  "Q8b",  "Q8c",  "Q8d",  "Q8e",  "Q8f", "Q8g",  "Q8h" , "Q8i" , "Q8j" , "
  Q5 = c(rep("Q5", 17))
)

env_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
  Significant = logical(),
  Test = character()
)

for (i in seq_len(nrow(env_dic))) {

  env_results[i, ] <- validation(
    question1 = env_dic$q_ref[i],
    question2 = env_dic$Q5[i],
    data = m_coalesced
  )
}

env_results <- env_results %>%
  mutate(
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
    Sig_Adj = Adj_P <= 0.05,
    Question1 = env_dic$q_ref,
    Question2 = env_dic$Q5
  ) %>%
  arrange(Adj_P)

env_results
dentists <- m_coalesced %>%
  filter(role == "Dentist")
```

```r
prac_dic <- data.frame(
  q_ref = c("Q8a",  "Q8b",  "Q8c",  "Q8d",  "Q8e",  "Q8f", "Q8g",  "Q8h" , "Q8i" , "Q8j" , "(
  Q3 = c(rep("Q3", 18))
)

prac_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
  Significant = logical(),
  Test = character()
)

for (i in seq_len(nrow(prac_dic))) {

  prac_results[i, ] <- validation(
    question1 = prac_dic$q_ref[i],
    question2 = prac_dic$Q3[i],
    data = dentists
  )
}

prac_results <- prac_results %>%
  mutate(
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
    Sig_Adj = Adj_P <= 0.05,
    Question1 = prac_dic$q_ref,
    Question2 = prac_dic$Q3
  ) %>%
  arrange(Adj_P)

prac_results

addt_dic <- data.frame(
  q_ref = c("Q8a",  "Q8b",  "Q8c",  "Q8d",  "Q8e",  "Q8f", "Q8g",  "Q8h" , "Q8i" , "Q8j" , "(
  q_group = c(rep("Q12", 14), "Q9", "Q9", "Q13", "Q13")
)

addt_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
```

```
  Significant = logical(),
  Test = character()
)

for (i in seq_len(nrow(addt_dic))) {

  addt_results[i, ] <- validation(
    question1 = addt_dic$q_ref[i],
    question2 = addt_dic$q_group[i],
    data = m_coalesced
  )
}

addt_results <- addt_results %>%
  mutate(
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
    Sig_Adj = Adj_P <= 0.05,
    Question1 = addt_dic$q_ref,
    Question2 = addt_dic$q_group
  ) %>%
  arrange(Adj_P)

addt_results
stack_bar <- function(q_reference, q_group, data){
  bar_df <- data %>%
    group_by(.data[[q_reference]], .data[[q_group]]) %>%
    summarize(counts = n(), .groups = "drop") %>%
    drop_na() %>%
    group_by(.data[[q_group]]) %>%
    mutate(percent = counts / sum(counts) * 100) %>%
    ungroup()

  p <- ggplot(bar_df, aes(x = .data[[q_reference]], y = percent, fill = .data[[q_reference]])
    geom_col() +
    facet_wrap(vars(.data[[q_group]])) +
    labs(
      title = paste(q_reference, "vs", q_group, sep = " "),
      x = NULL,
      y = "Percent of responses"
    ) +
    theme(
      axis.text.x = element_blank(),
```

```
      axis.ticks.x = element_blank()
    )
  return(p)
}
plots_comfort <- pmap(comfort_dic,
                      function(q_ref, role){
                        stack_bar(
                          q_reference = q_ref,
                          q_group = role,
                          data = m_coalesced
                        )
                      })
walk(plots_comfort, print)
plots_knowledge <- pmap(knowledge_dic,
                        function(q_ref, role){
                          stack_bar(
                            q_reference = q_ref,
                            q_group = role,
                            data = m_coalesced
                          )
                        })
walk(plots_knowledge, print)
plots_quiz <- pmap(quiz_dic,
                        function(q_ref, role){
                          stack_bar(
                            q_reference = q_ref,
                            q_group = role,
                            data = m_coalesced
                          )
                        })
walk(plots_quiz, print)
plots_env <- pmap(env_dic,
                        function(q_ref, Q5){
                          stack_bar(
                            q_reference = q_ref,
                            q_group = Q5,
                            data = m_coalesced
                          )
                        })
walk(plots_env, print)
plots_dentists <- pmap(prac_dic,
                        function(q_ref, Q3){
```

```
                        stack_bar(
                          q_reference = q_ref,
                          q_group = Q3,
                          data = dentists
                        )
                      })
walk(plots_dentists, print)
plots_addt <- pmap(addt_dic,
                      function(q_ref, q_group){
                        stack_bar(
                          q_reference = q_ref,
                          q_group = q_group,
                          data = m_coalesced
                        )
                      })
walk(plots_addt, print)
```